



Kmeans

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U.S. DEPARTMENT
of **ENERGY**

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Activating RSA SecurID Token

1. Initiate an ssh connection to `username@home.ccs.ornl.gov`.
 - Type `ssh <csep##>@frontier.olcf.ornl.gov` in the terminal.
2. When prompted for a PASSCODE, enter the 6-digit **token code** displayed on your token.
3. When asked if you are ready to set your PIN, answer with 'y' and enter a 4-8 digit alphanumeric of your choice (i.e. **123abc**). You will then be prompted, re-enter your PIN.
4. You will then be prompted to enter your PASSCODE.
 - **Wait until the next 6 digit code appears on your token and enter your PIN + token code.**
 - ex.) **123abc + 123456** (no spaces)
5. Your PIN is now set and your token activated and ready for use.

Please return your token + envelope at the end of the day.



IF YOU DO NOT HAVE
AN SSH CLIENT LIKE
POWERSHELL OR
TERMINAL, SET UP
YOUR TOKEN ON
ONE OF OUR
LAPTOPS.

Big Idea: Use Machine Learning to explore relationships between power outage frequency, location, and time of year.

Motivation: Understanding when and why power outages occur helps plan grid maintenance and operation.

Data: Eagle-I, a Department of Energy data base, managed from Oak Ridge National Lab, that contains data about the number of customers without power in any given location in the United States or its territories.

Our specific raw data: number of customer without of power every 15 minutes each each of the 3081 counties in the lower continental United States from 2014 to 2022.

How much data is that?

3081 counties * 4 sets of fifteen min per hour * 24 hours per day * 365.25 days per year * 9 years.

= 972,289,656 data points or 8 GB.

Big Data and MPI: Raw data

We have 8 GB of Egale-I Data from 2014-2022 for every county in the United States;
The data shows how many customer were without power every fifteen minutes of each day.

What do we actually want?

An OUTAGE EVENT

- Where?
- When did it start?
- When did it end?
- How long did it last?
- How large was it?
- What region?
- What time of year?

RAW DATA

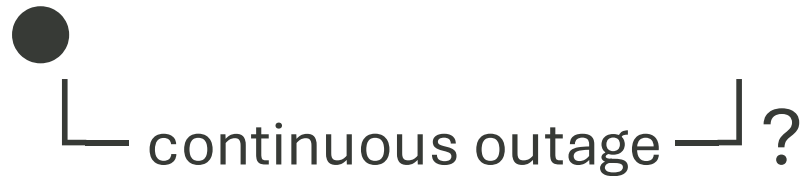
FIPS	customers out	timestamp
47093	12	2019-07-14 14:00
47093	143	2019-07-14 14:15
47093	161	2019-07-14 14:30
47093	155	2019-07-14 14:45
47093	2	2019-07-14 19 :00
...		

How long are the outages?

2:00 2:15 2:30 2:45



7:00 7:15



continuous outage ?

```
if next_time == current_time + 15 minutes:
    continue
else:
    # this outage has ended
```

RAW DATA

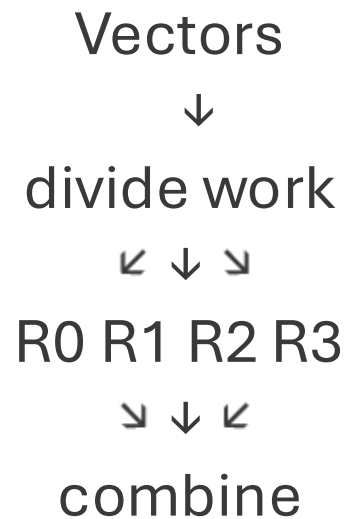
FIPS	customers out	timestamp
47093	12	2019-07-14 14:00
47093	143	2019-07-14 14:15
47093	161	2019-07-14 14:30
47093	155	2019-07-14 14:45
47093	10	2019-07-14 19:00
47093	8	2019-07-14 19:15

...

If you want to see the whole script and how it works look at the explanation in **Parts1and2Poweroutages.ipynb** here:
https://github.com/suzannepk/anatomy_of_a_power_outage

Where does MPI come in?

Dot Product example, recall:



Power outage data
8 GB outage observations



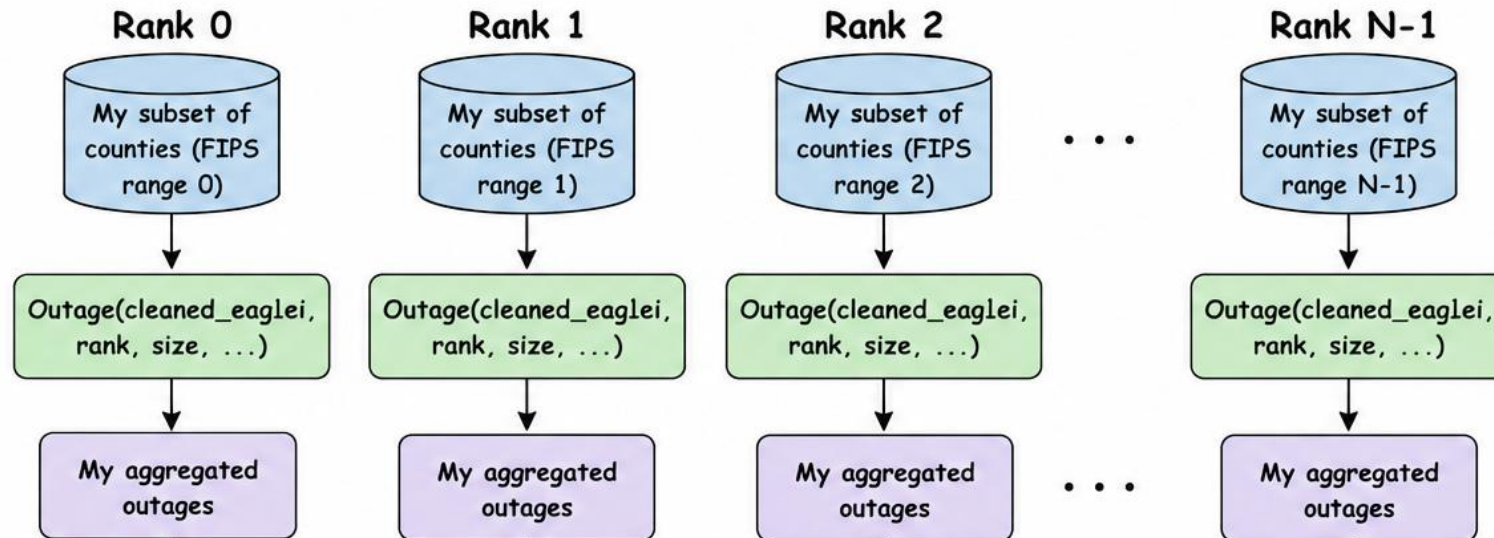
Different language. Different scientific problem. Same parallel-thinking pattern.

What this code is doing (big picture)

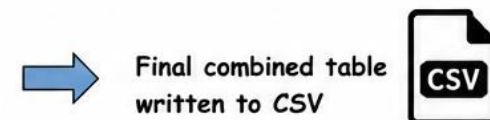
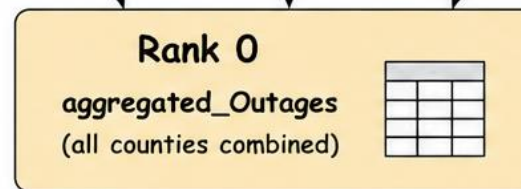
```
rank = comm.Get_rank()
size = comm.Get_size()

agg_outages_per_proc = Outage(cleaned_eaglei, rank, size, ...)
aggregated_Outages = comm.gather(agg_outages_per_proc, root=0)
```

1. Each MPI process works on its portion of the data



2. gather() collects all results on root (Rank 0)



Let us all pretend that we are going to do a scaling study
Time the data cleaning program when it runs on the following
number of cores:

#cores	#time to completion
---------------	----------------------------

1	
----------	--

2	
----------	--

4	
----------	--

8	
----------	--

16	
-----------	--

32	
-----------	--

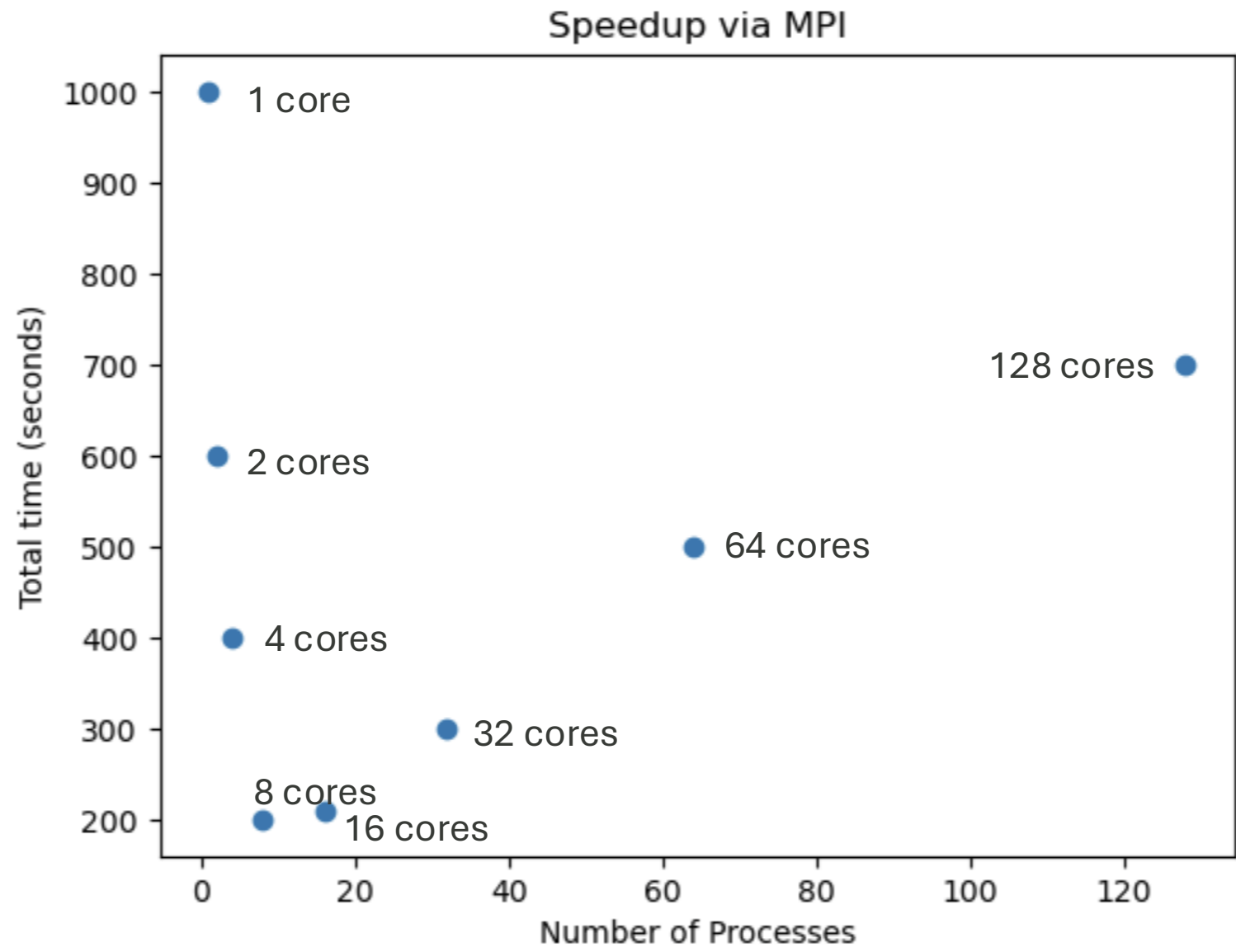
64	
-----------	--

128	
------------	--

Let us all pretend that we are going to do a scaling study
Time the data cleaning program when it runs on the following
number of cores:

#cores	#time to completion
1	1000s
2	598s
4	410s
8	200s
16	218s
32	300s
64	500s
128	700s

What do we gain?



Does having multiple processors make cleaning and processing the data faster?

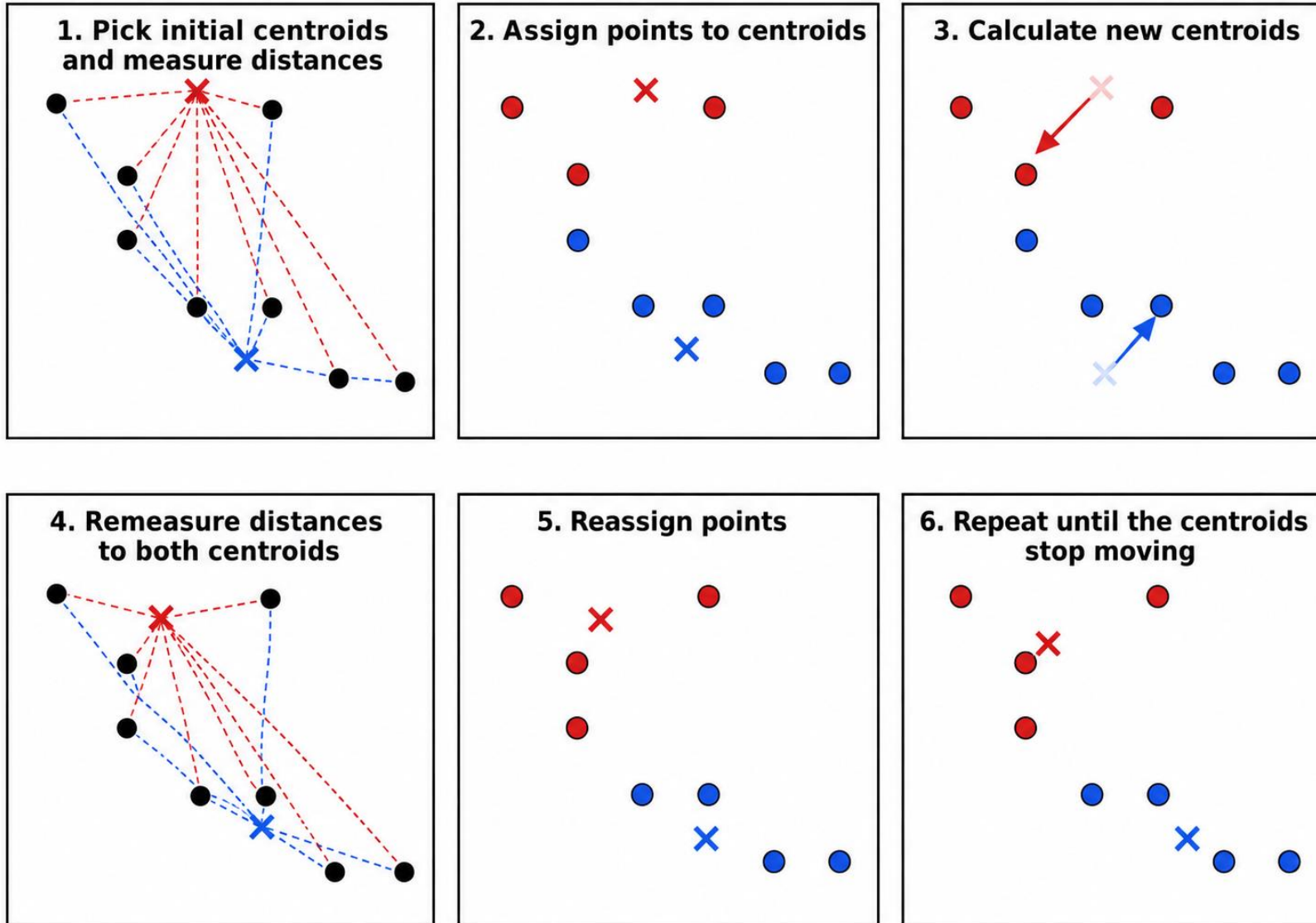
Why does the time to complete the job get slower when using more than 8 cores?

What we have learned so far

- Preparing data for ML or AI requires understanding the provenance of the data and its associated metadata.
- Data must be cleaned and processed into the form best suited to the questions you want your ML algorithm to answer.
- MPI can help you efficiently process operations that need to be performed repeatedly across large datasets. However, you must also account for the cost of communication between processes.

Now let's talk about K-means clustering, the Unsupervised ML that we will use to cluster our power outages by locations and time of year.

Kmeans



This is unsupervised machine learning because we do not give the algorithm pre-labeled examples.

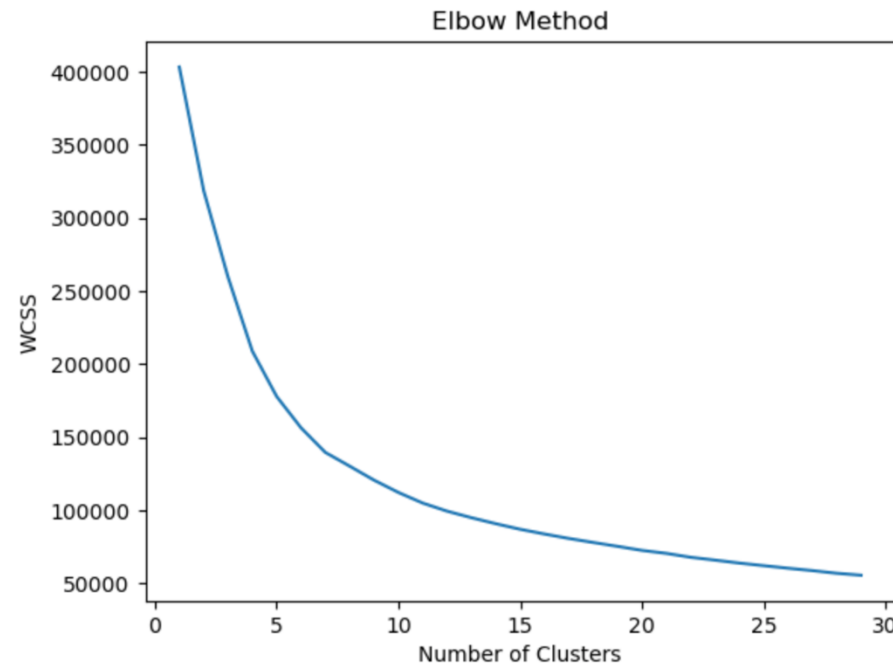
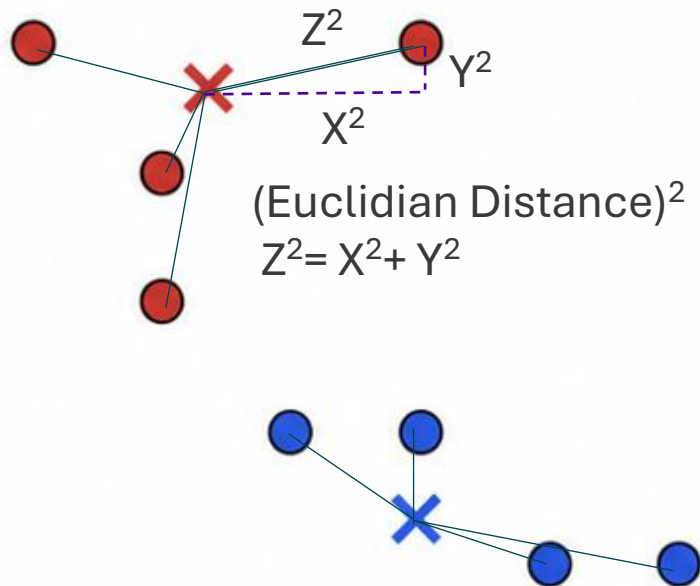
K-means groups points by similarity, based on distance to the cluster centroids.

Kmeans: Choosing the number of clusters

Set K to 2, converge Kmeans, find WCSS, Set K to 3, converge Kmeans, find WCSS, Set K to 4, converge Kmeans, find . . .

You pick the K, that is just before the point where having more clusters doesn't get you much more tightening– the Elbow

Summary



”The Elbow Method”- helps you choose the number of clusters for a set of points.

Within Cluster Sum of Squares (WCSS) – sum of the square of the Euclidian distances between the points and their centroid in each cluster. Measure of how tightly clustered the points are in each group.

The magnitude of X and Y variable must be of the same size scale or normalized so the Euclidian distances are not skewed to the large scale.

Choose K based on the elbow of the plot.

This is only one of three methods you will use for choosing K.

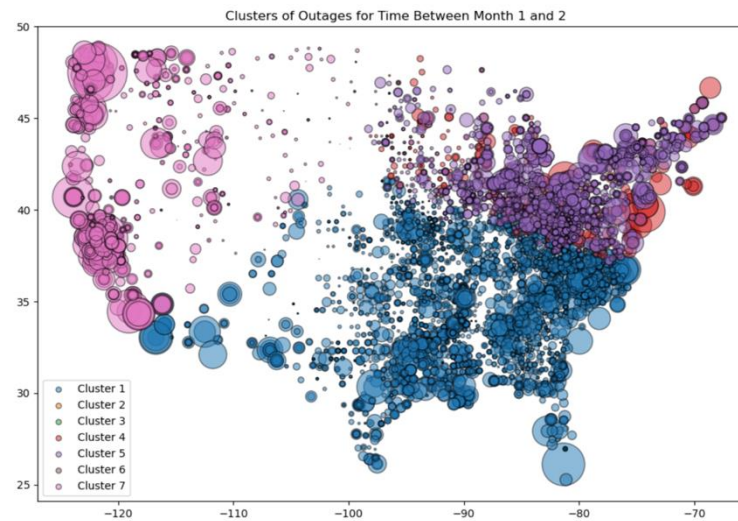
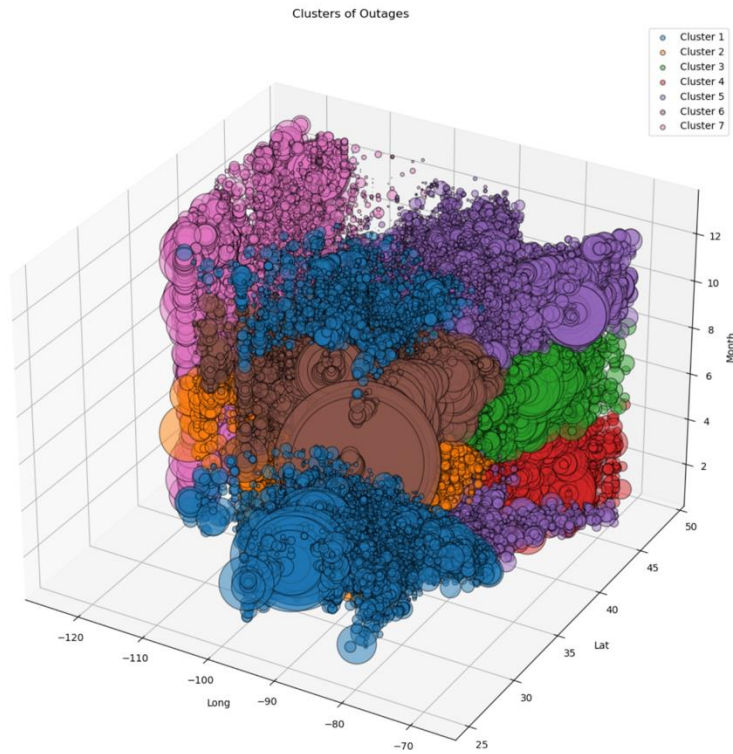
What we will do during hands-on.

Use cleaned data to plot all the outages that happened from 2014-2022 on on plot. Lat, long Month,

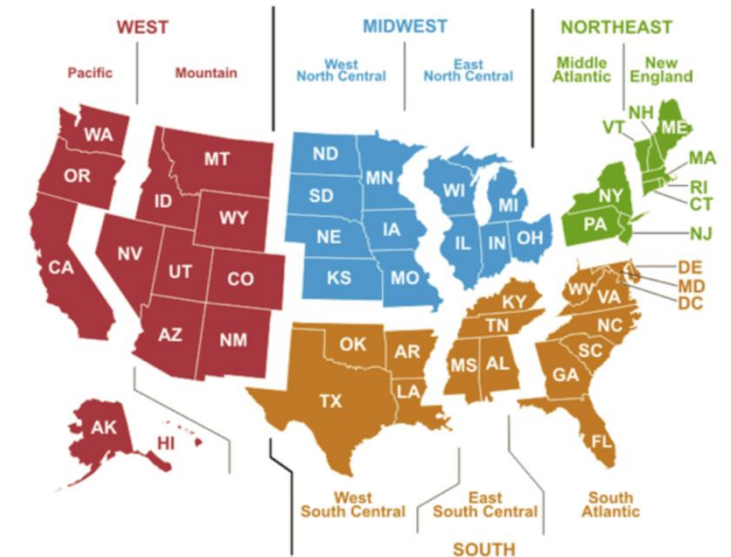
Use K-mean to group them– you will need to run tests and determine how many groups will be ideal.

What do the groups tell you about time of year different regions of the US experience power outages?

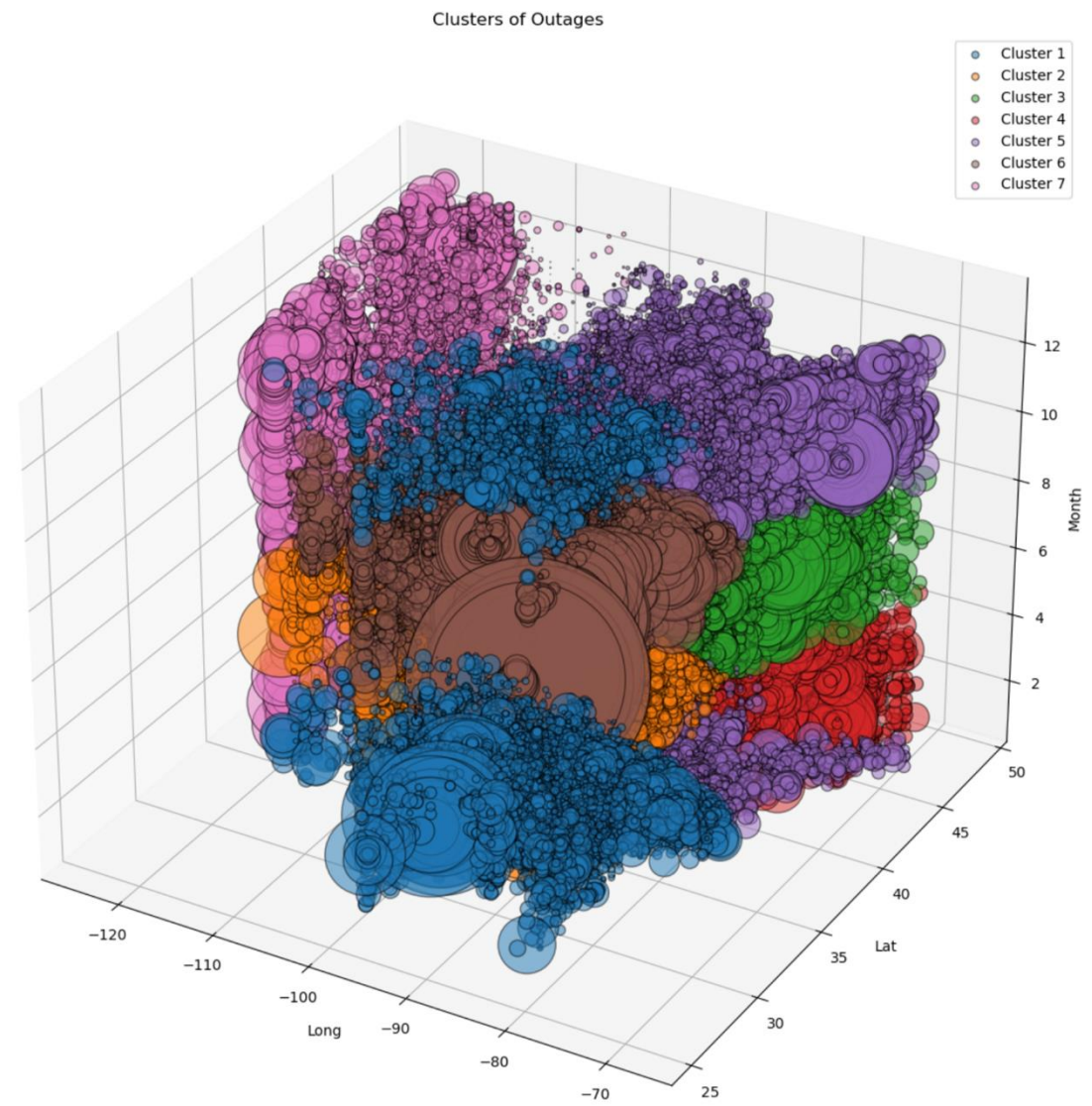
Is there weather that can start to help explain this? What else besides weather could contribute?



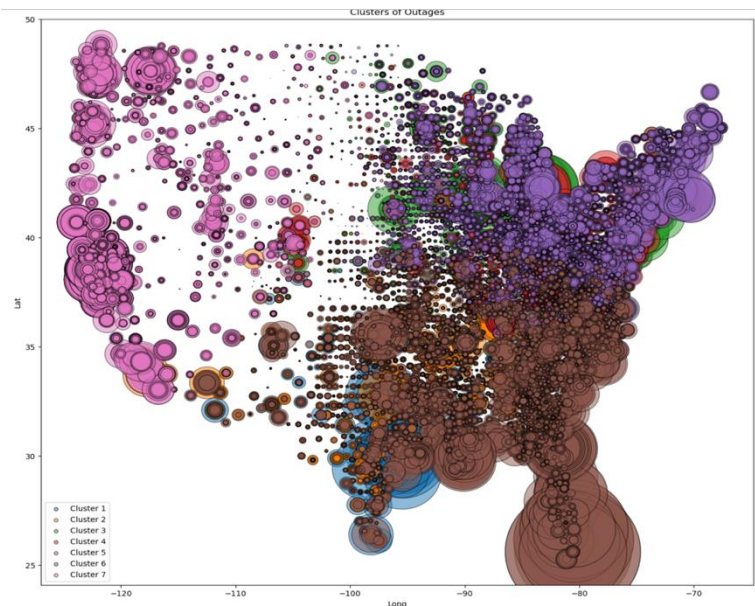
Top view



Data is spread vertically over the the months of the year for 2014-2022. What pattern do you notice?

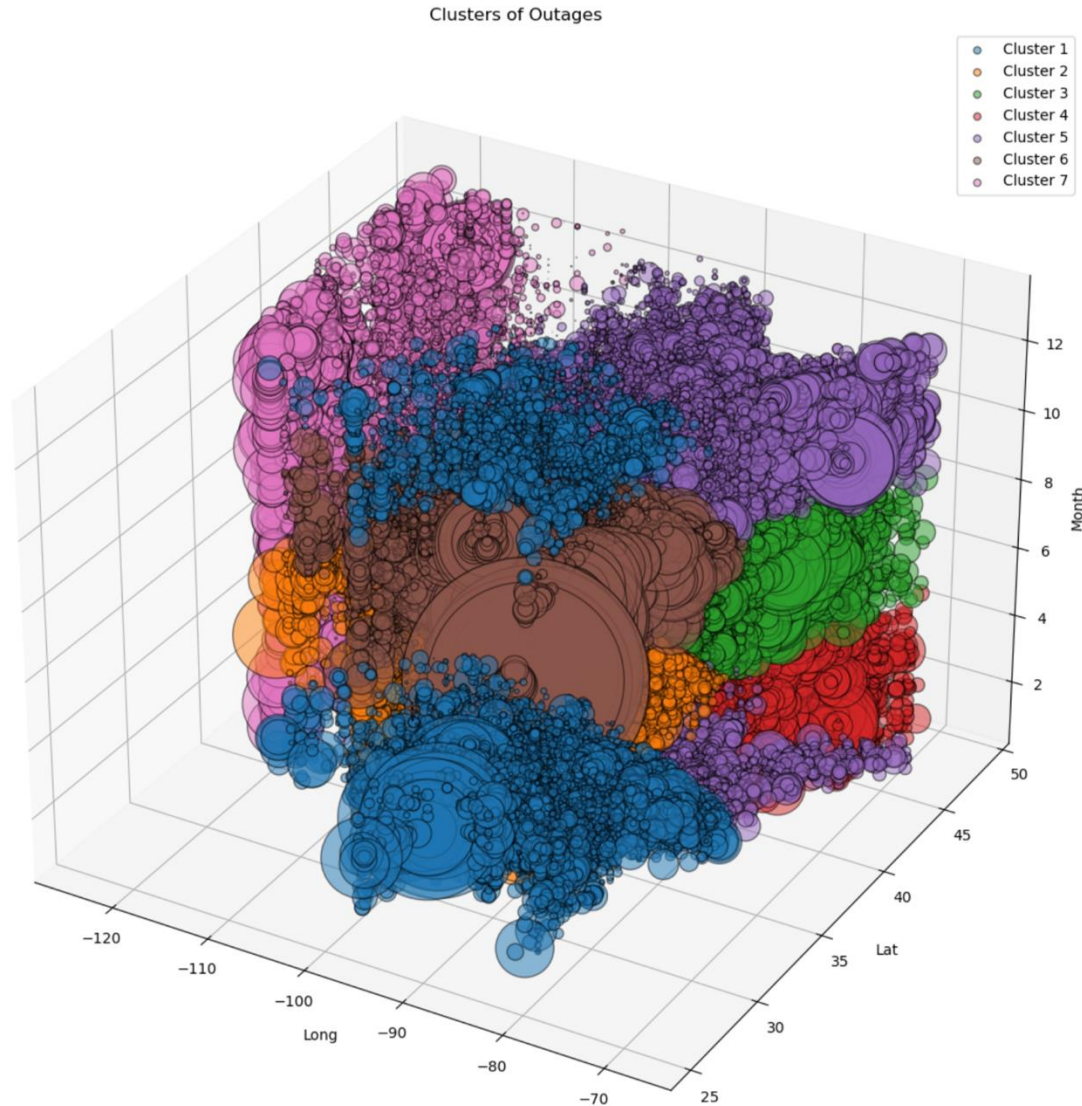


Why do you suppose the blue and purple groups appear in both month 1 and 12?



September slice

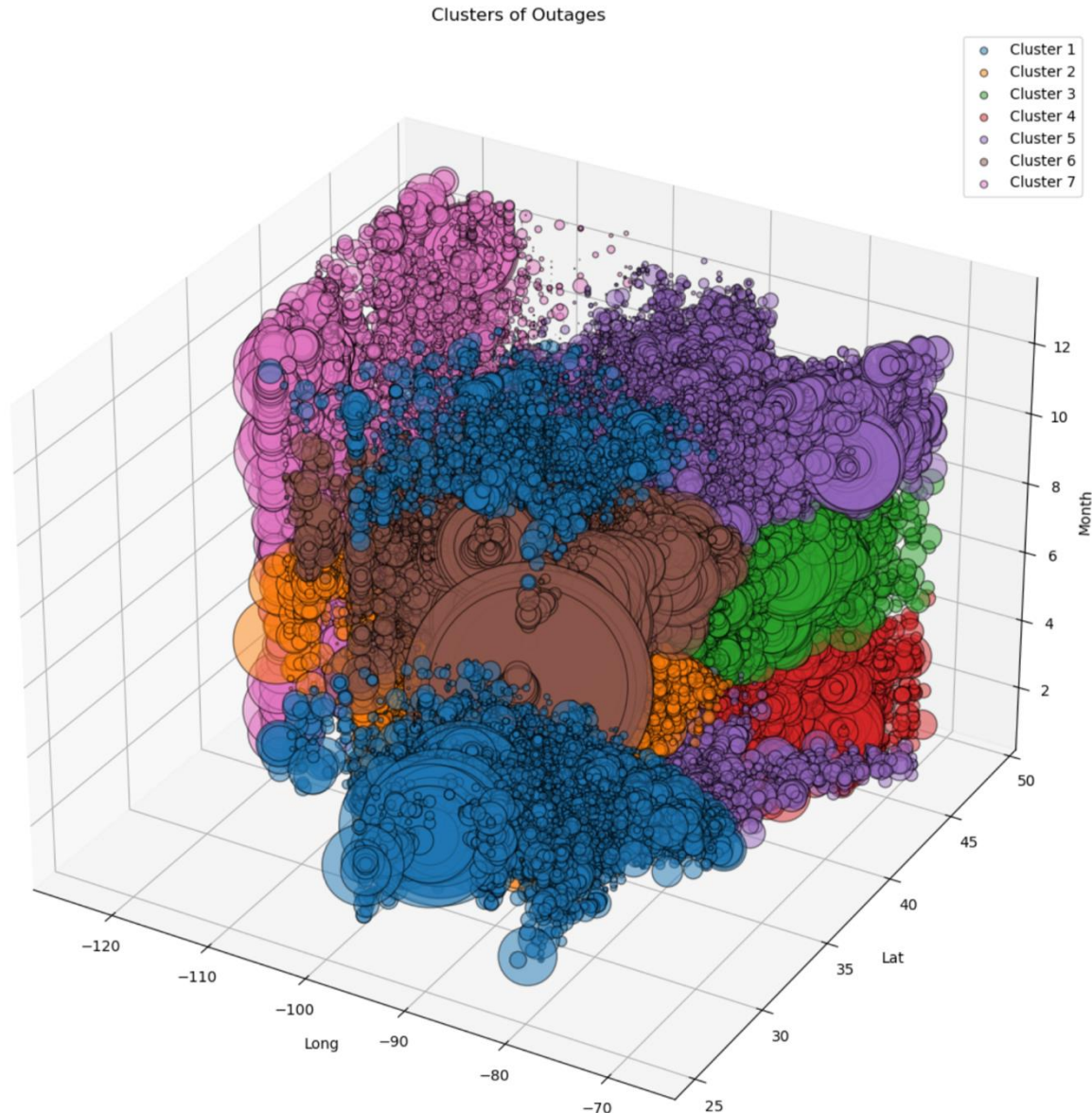
Data is spread vertically over the the months of the year for 2014-2022. What pattern do you notice?



Why do you suppose The blue and purple groups appear in both month 1 and 12?

Reminder: For the K-means grouping, we used normalized coordinates. For time of year, we also used cyclic coordinates. However, for plotting, we use the unnormalized data points in their original units.

K=7 with Month- so data is spread over the the months of the year. What pattern do you notice?



Centroid for Cluster 1 occurs during time of year: 1.7111280047380604

Centroid for Cluster 2 occurs during time of year: 5.735178050639296

Centroid for Cluster 3 occurs during time of year: 7.405455893723385

Centroid for Cluster 4 occurs during time of year: 3.6990201263266385

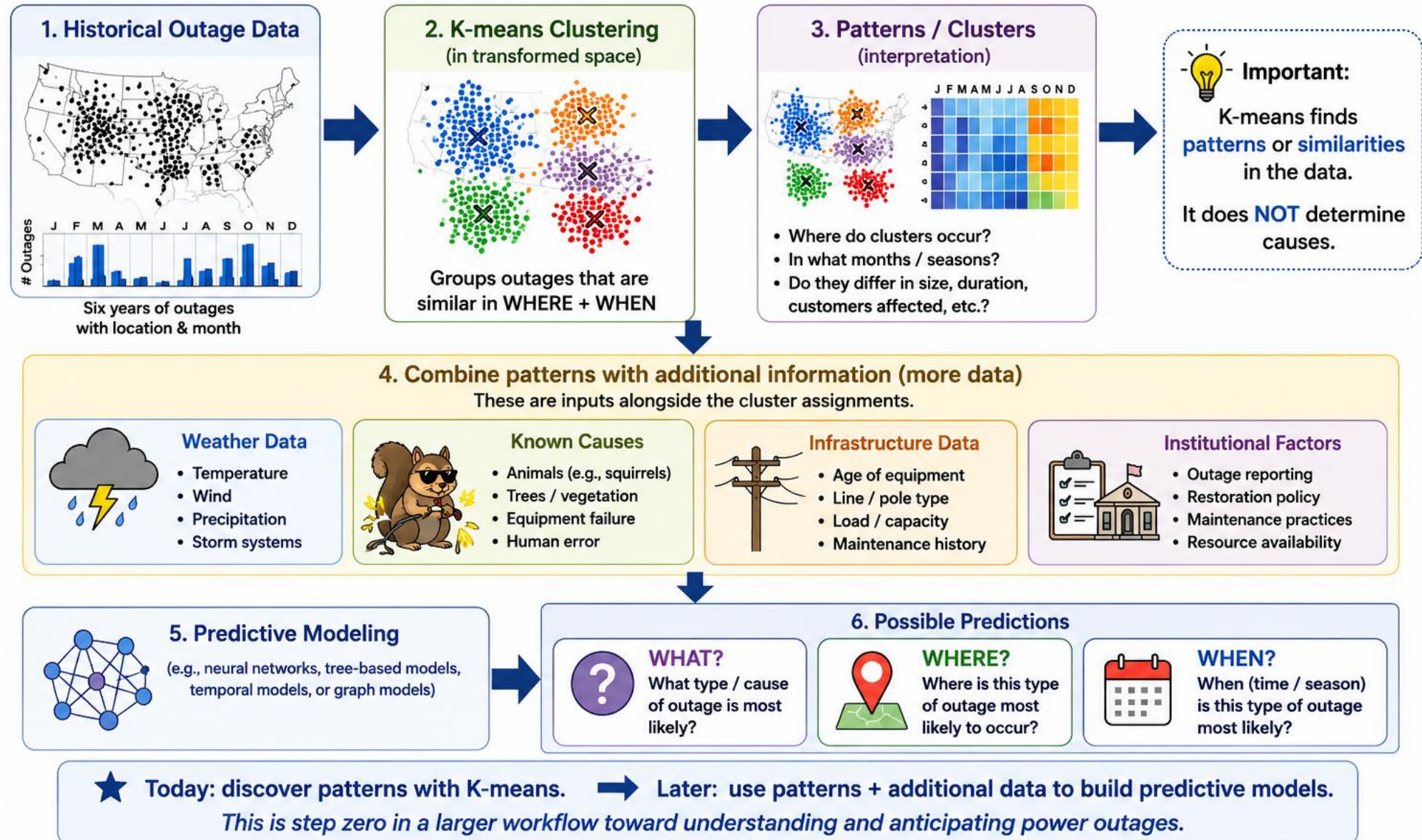
Centroid for Cluster 5 occurs during time of year: 11.496448998781853

Centroid for Cluster 6 occurs during time of year: 9.45702559695134

Centroid for Cluster 7 occurs during time of year: 10.33476853653173

K-means is Step Zero: From Patterns to Prediction

Today we discover patterns. Later we combine them with more data to make predictions.



Jupyter

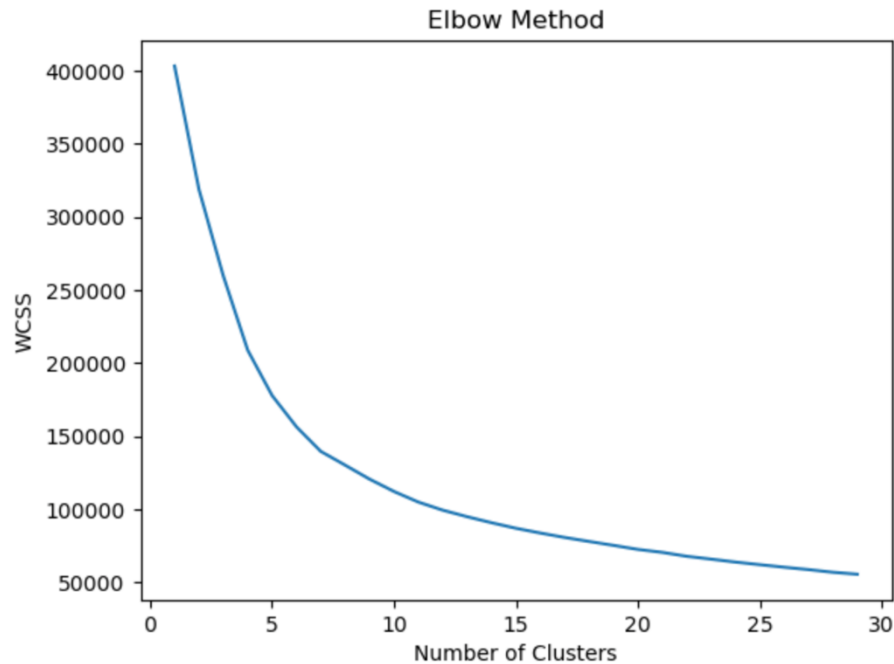
How to Run This Lab

- Clone this repository on OLCF Jupyter.
- Open a browser and go to <https://jupyter.olcf.ornl.gov/>
- Log in with your csep*** username and PIN + token code.
- Select the GiMS instance. 
- Once it starts, click on the Terminal icon.
- From the terminal, cut and paste the following command:
git clone https://github.com/suzannepk/K_means.git
hit enter
- You should see something like this:

```
[62956@jupyter-suzanne ~]$ git clone https://github.com/suzannepk/K_means.git
Cloning into 'K_means'...
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
Receiving objects: 100% (12/12), 12.34 MiB | 26.05 MiB/s, done.
```
- In the left-column menu, click on K_means and then K-Means.ipynb to open the notebook.
- Follow the directions in the notebook to complete the challenge.
- The last step will ask you to fill out this Google Form, <https://forms.gle/SkC54qf6H8Gati6B7> with your observations from the challenge.

Canned Data In the event of internet issues

How many Clusters?



The best K is at the elbow of the plot

Davies-Bouldin Index

- Lowest number indicates probable best K value

2 clusters the Davies-Bouldin Score is: 1.8483834191477473

3 clusters the Davies-Bouldin Score is: 1.5358534408158608

4 clusters the Davies-Bouldin Score is: 1.2458227809350584

5 clusters the Davies-Bouldin Score is: 1.1479118886959347

6 clusters the Davies-Bouldin Score is: 1.1133245176080968

7 clusters the Davies-Bouldin Score is: 1.0196317837613542

8 clusters the Davies-Bouldin Score is: 1.1268161379179344

9 clusters the Davies-Bouldin Score is: 1.12863564515673

10 clusters the Davies-Bouldin Score is: 1.1746757717976988

11 clusters the Davies-Bouldin Score is: 1.190109365981992

Calinski-Harabasz Index – higher score indicates better defined clusters

2 clusters the Calinski-Harabasz Score is: 26289.75904904237

3 clusters the Calinski-Harabasz Score is: 26779.50914188052

4 clusters the Calinski-Harabasz Score is: 31223.414524468662

5 clusters the Calinski-Harabasz Score is: 31872.387076521387

6 clusters the Calinski-Harabasz Score is: 31749.217176264447

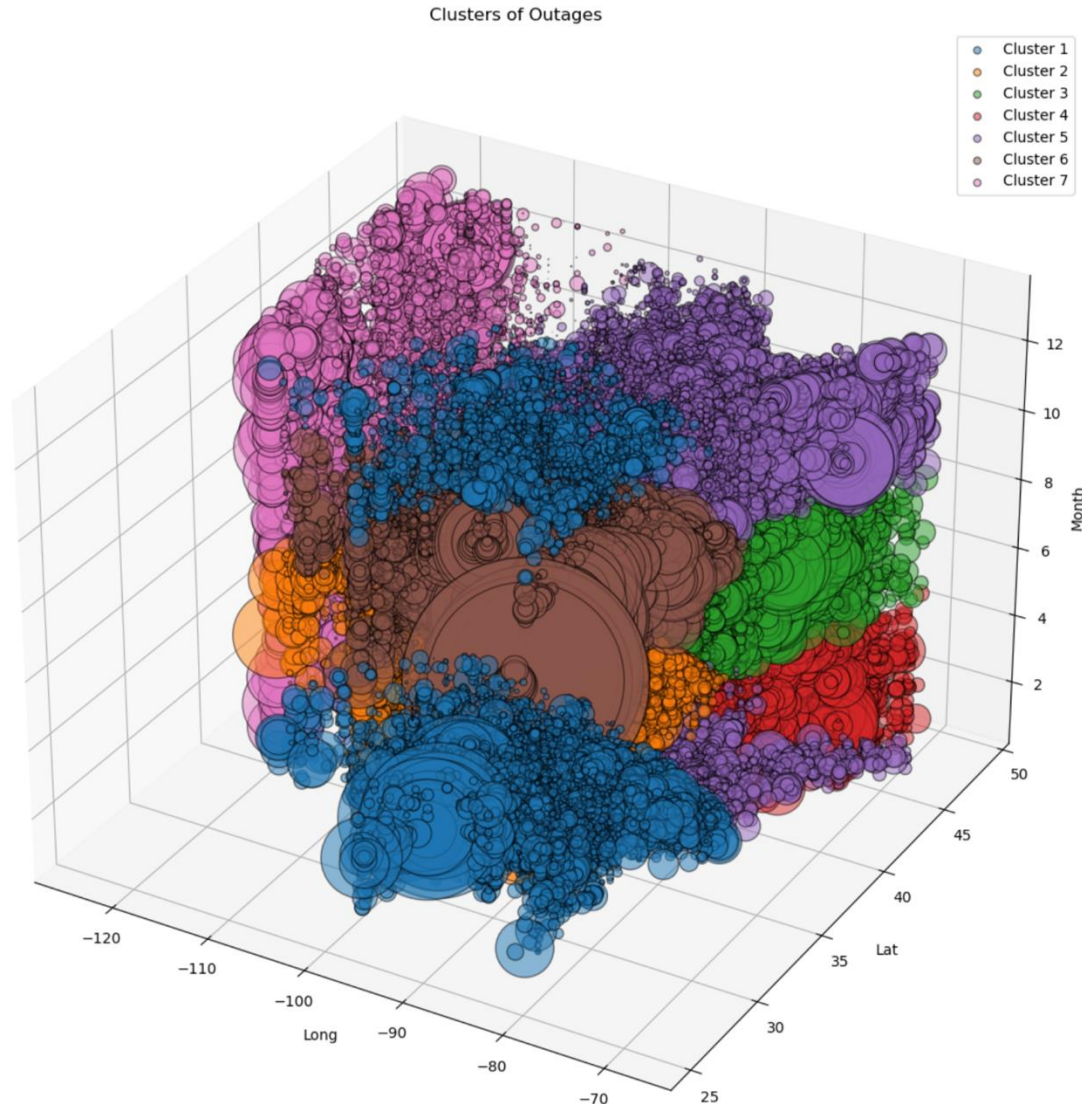
7 clusters the Calinski-Harabasz Score is: 31660.494254516598

8 clusters the Calinski-Harabasz Score is: 30135.44713020843

9 clusters the Calinski-Harabasz Score is: 29029.54135615725

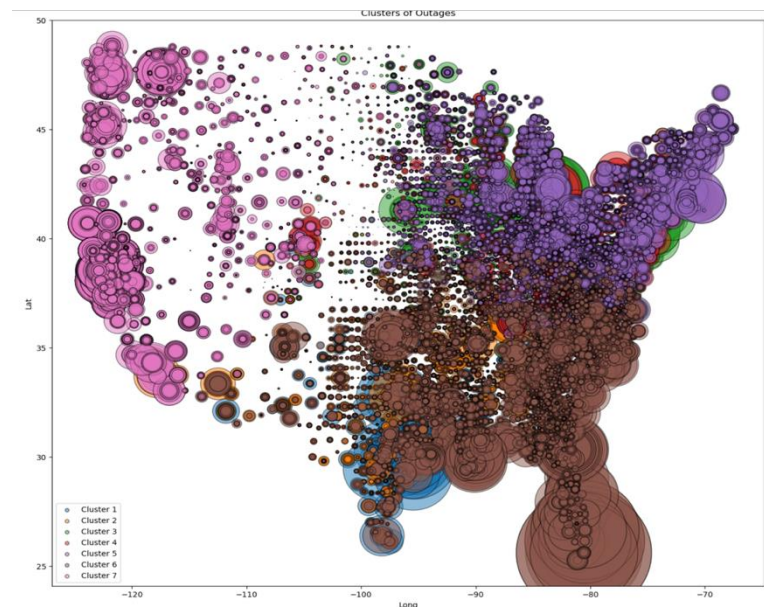
10 clusters the Calinski-Harabasz Score is: 28511.13392882875

K=7 with data is spread vertically over the the months of the year for 2014-2022. What pattern do you notice?



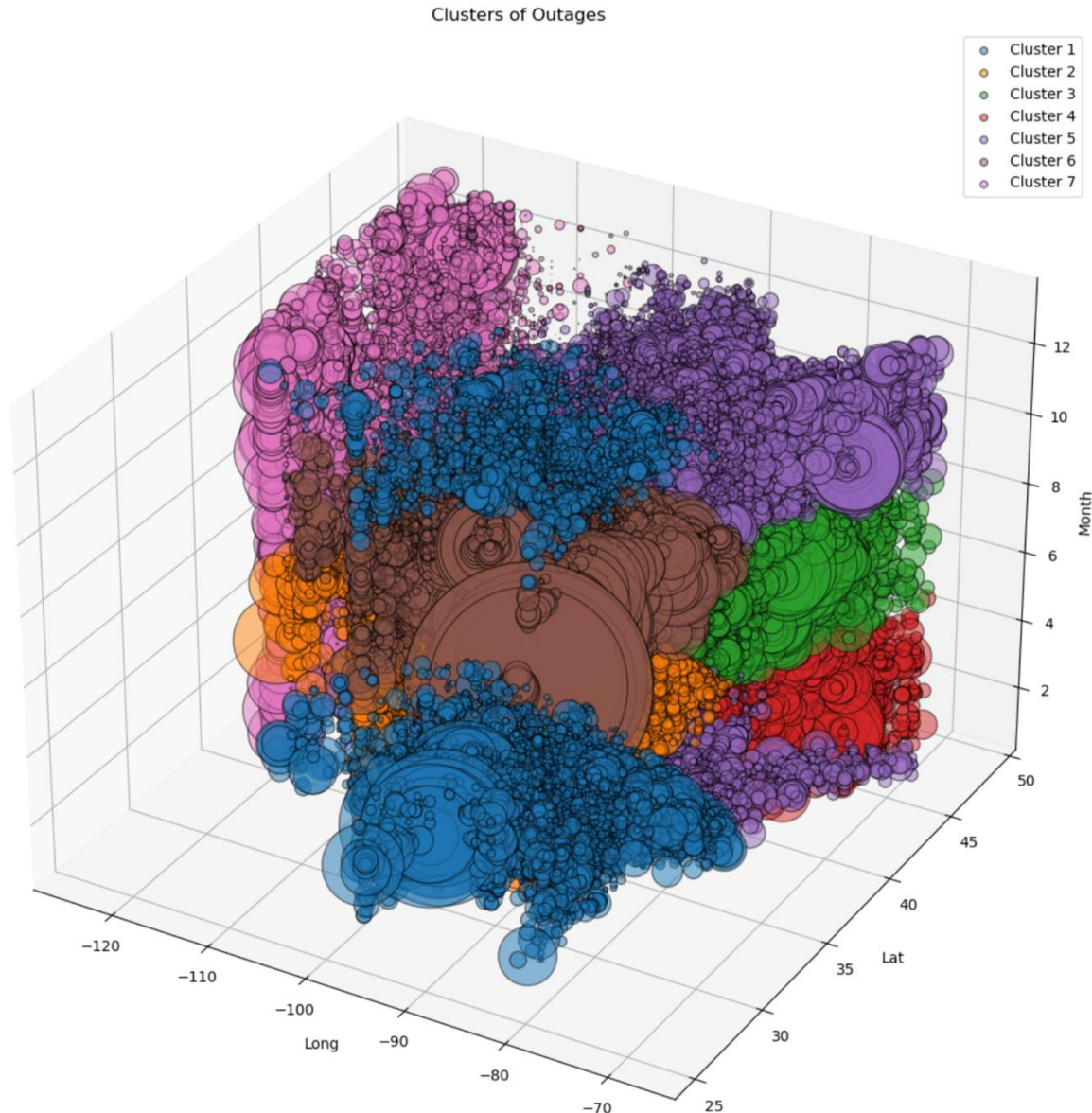
Why do you suppose The blue and purple groups appear in both month 1 and 12?

What weather might you guess is driving power outage in the warmer months? Would this still be true for arid regions?



Top view

K=7 with Month- so data is spread over the the months of the year. What pattern do you notice?



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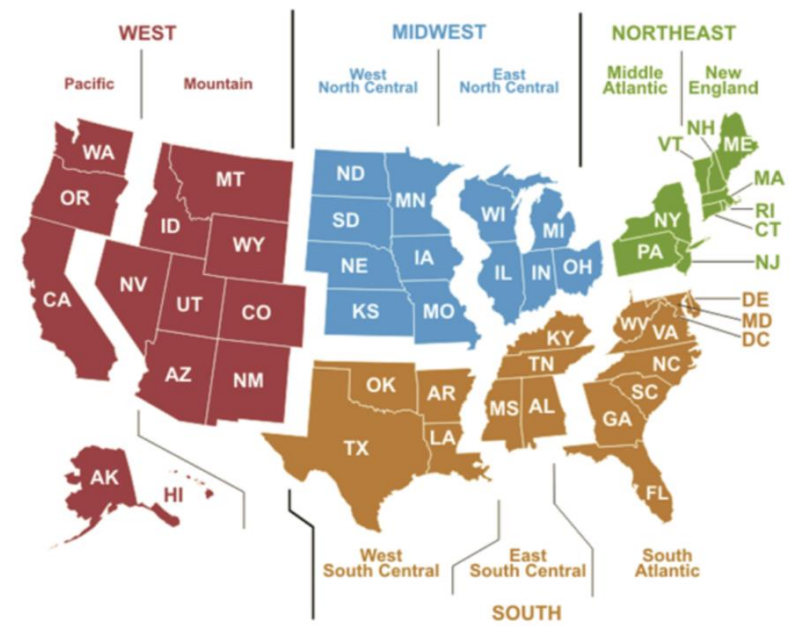
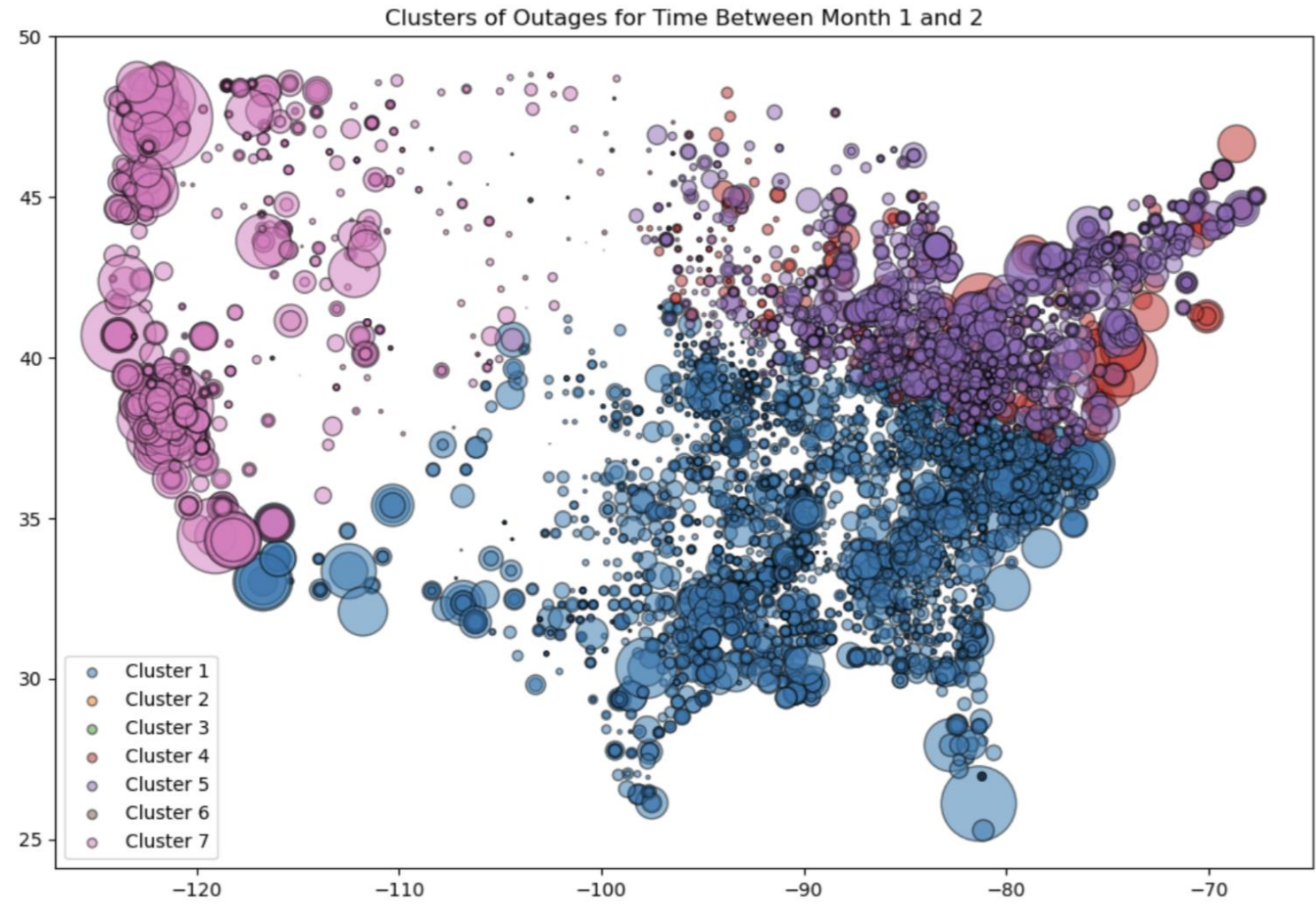
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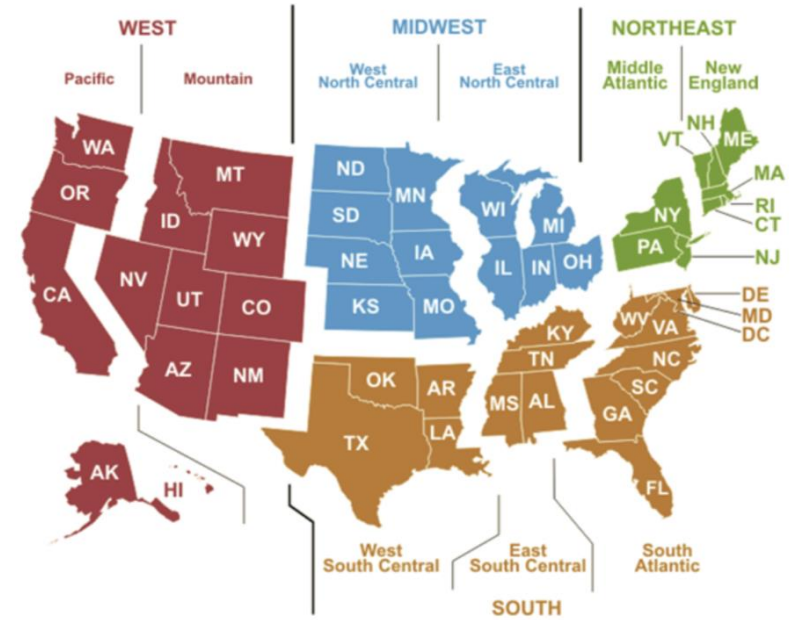
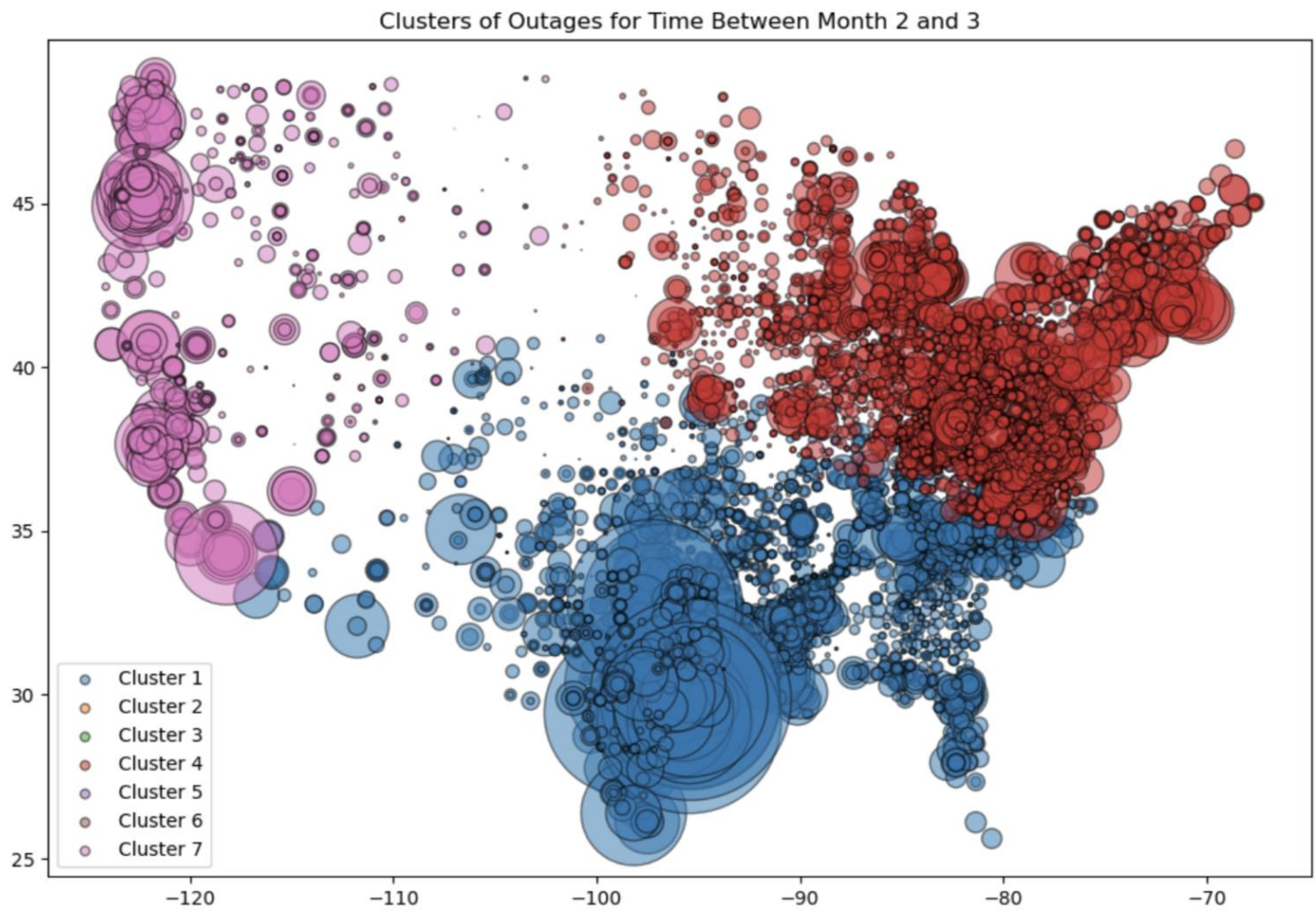
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K=7 with Moth- so data is spread over the the months of the year. What pattern do you notice?



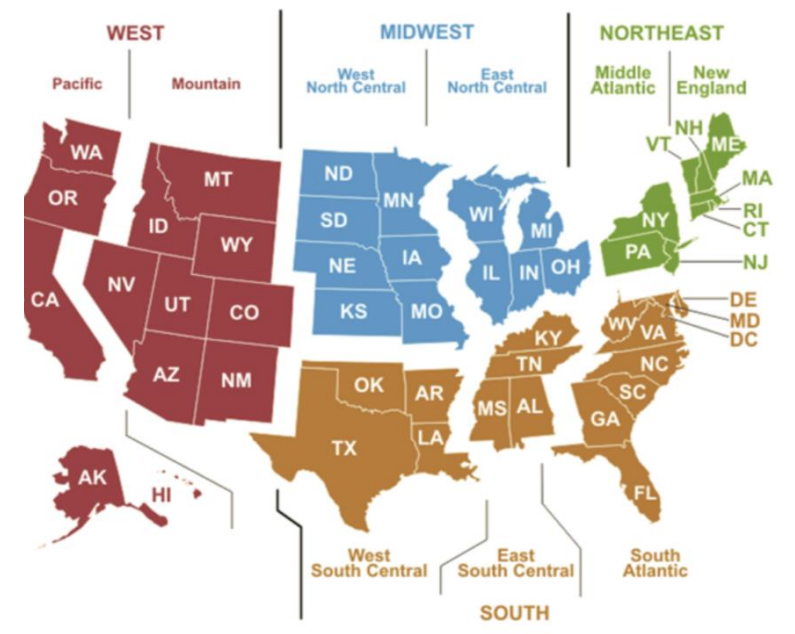
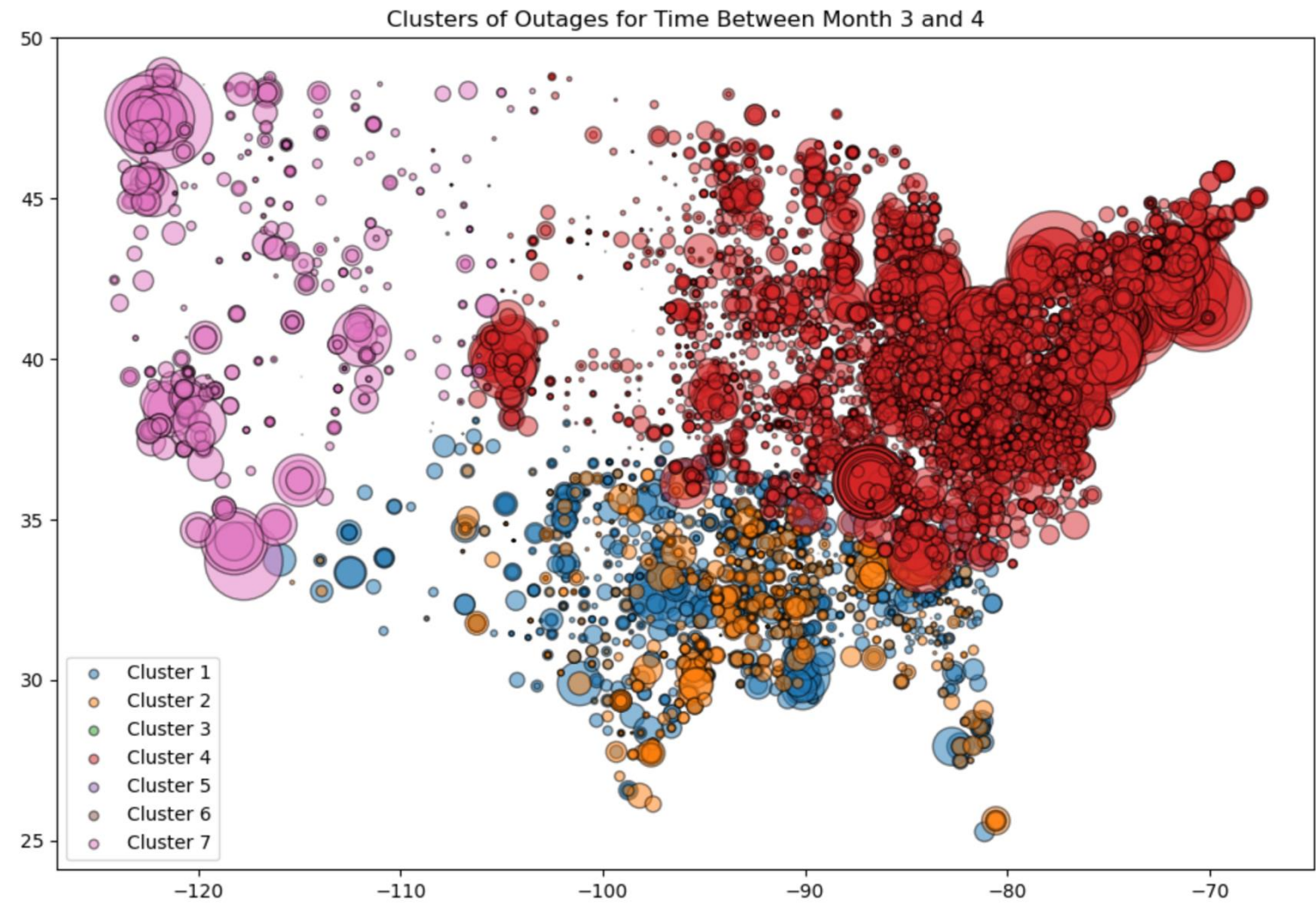
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<https://www.ncei.noaa.gov/news/great-texas-freeze-february-2021>

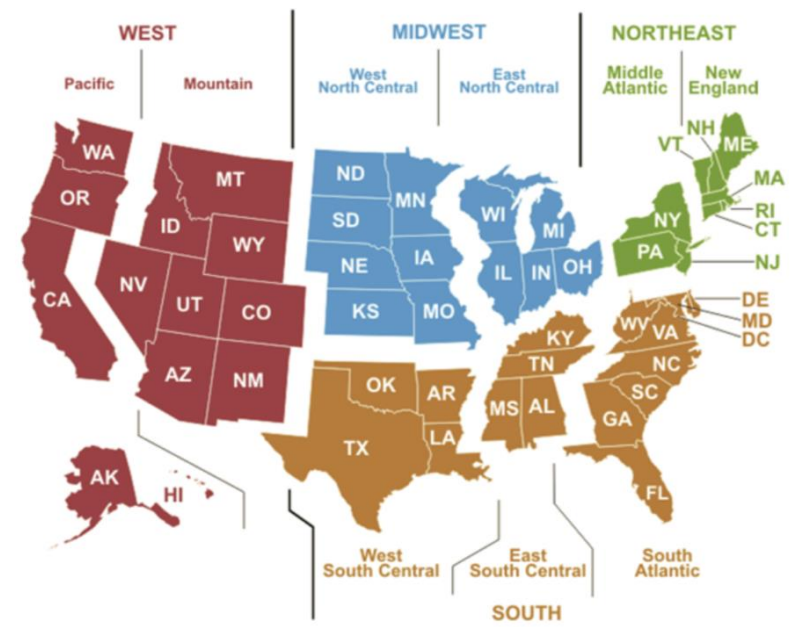
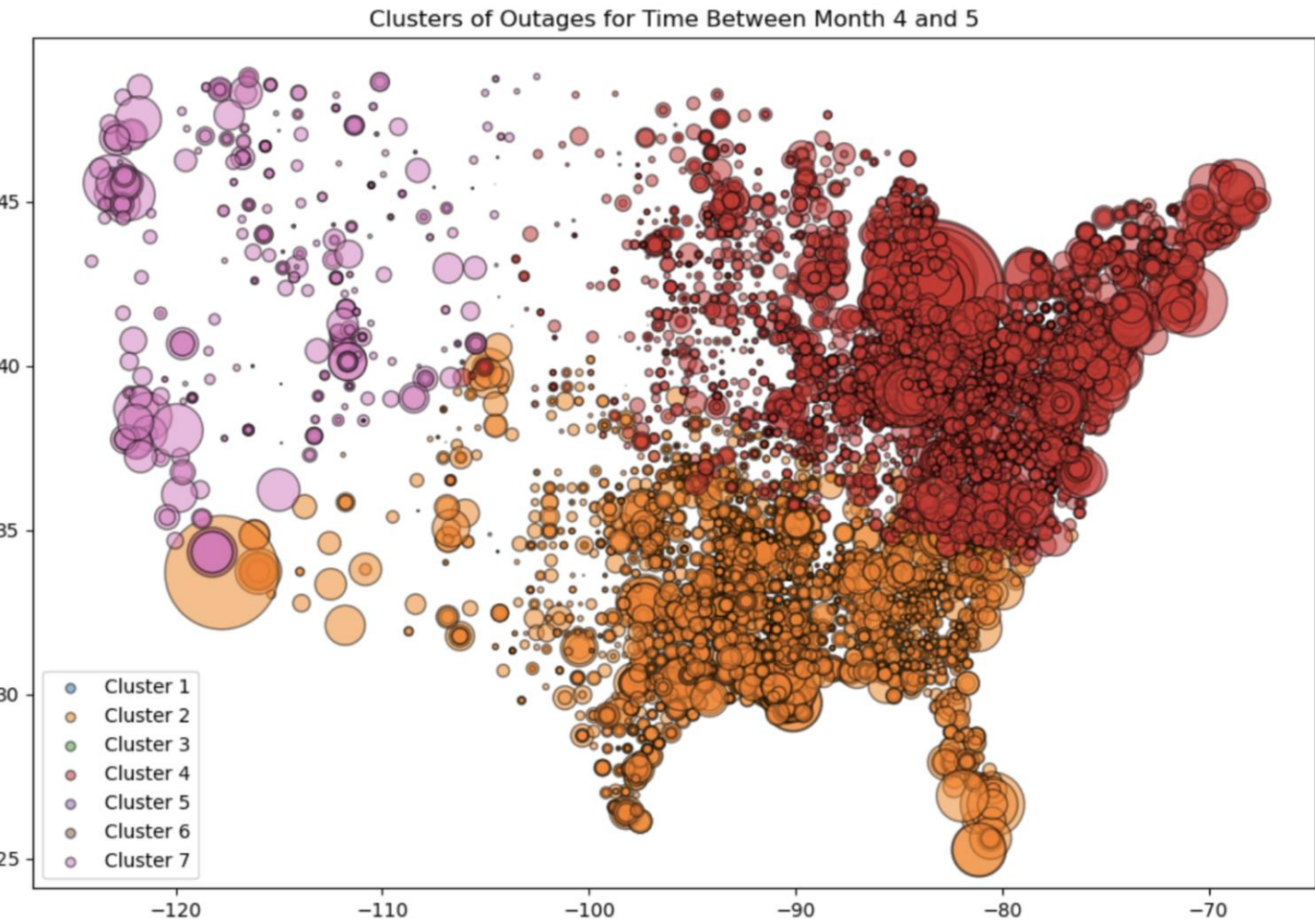
On February 11-20, 2021, a historical winter event occurred that set many records, including being the first billion-dollar weather disaster of that year. Though power outages occurred throughout the South, they were mostly concentrated in Texas. At the peak of the outage, nearly 10 million people were in the dark, lacking warmth and the ability to cook food.

K=7 with Moth- so data is spread over the the months of the year. What pattern do you notice?



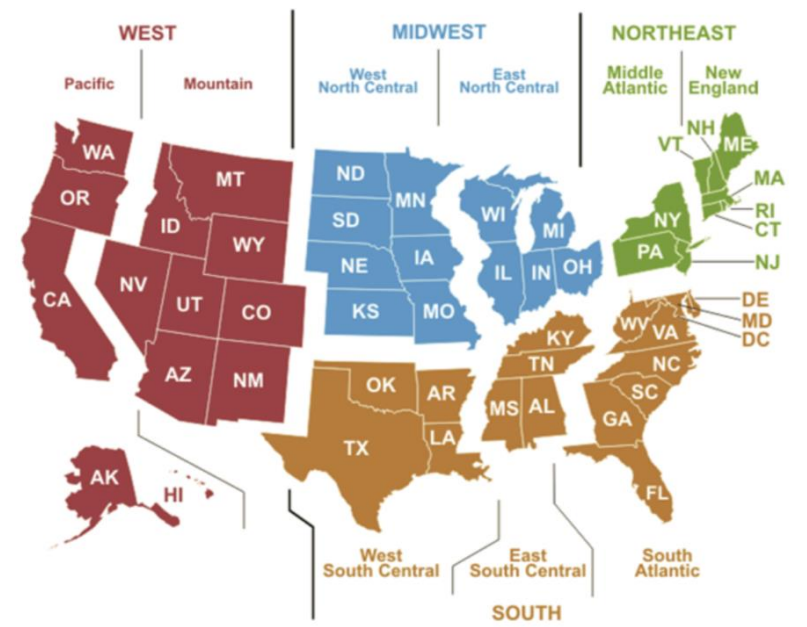
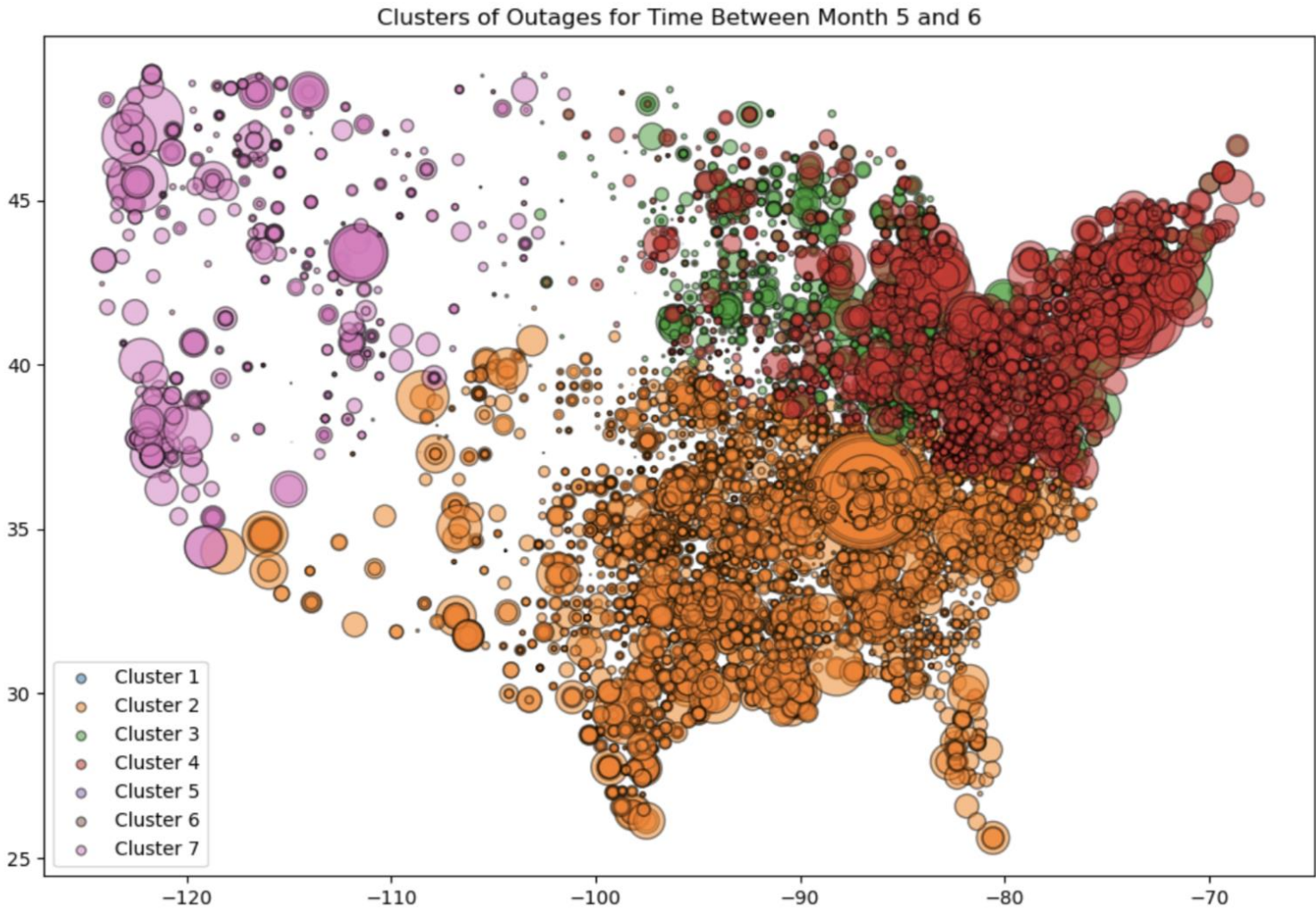
Regions 1: Pacific 2: Mountain 3: West North Central 4: West South Central 5: East North Central
6: East South Central 7: New England 8: Mid-Atlantic 9: South Atlantic

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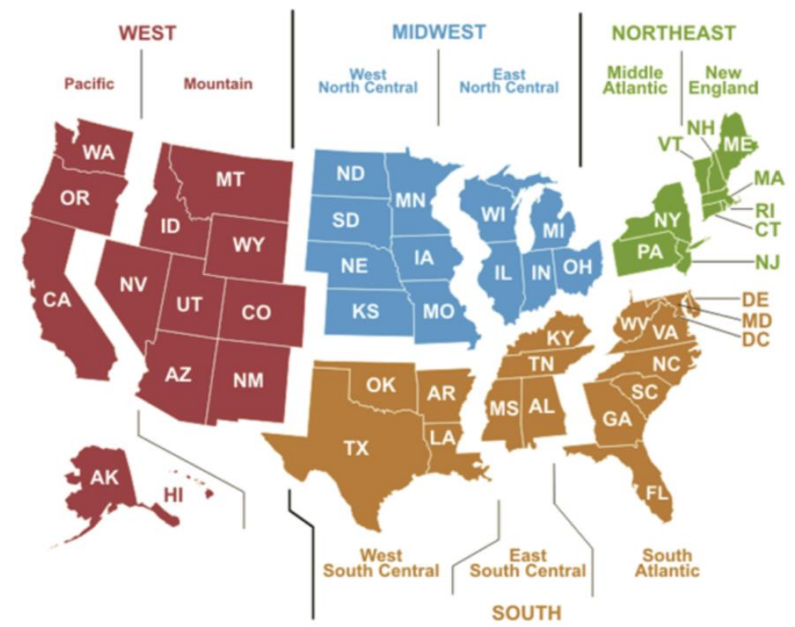
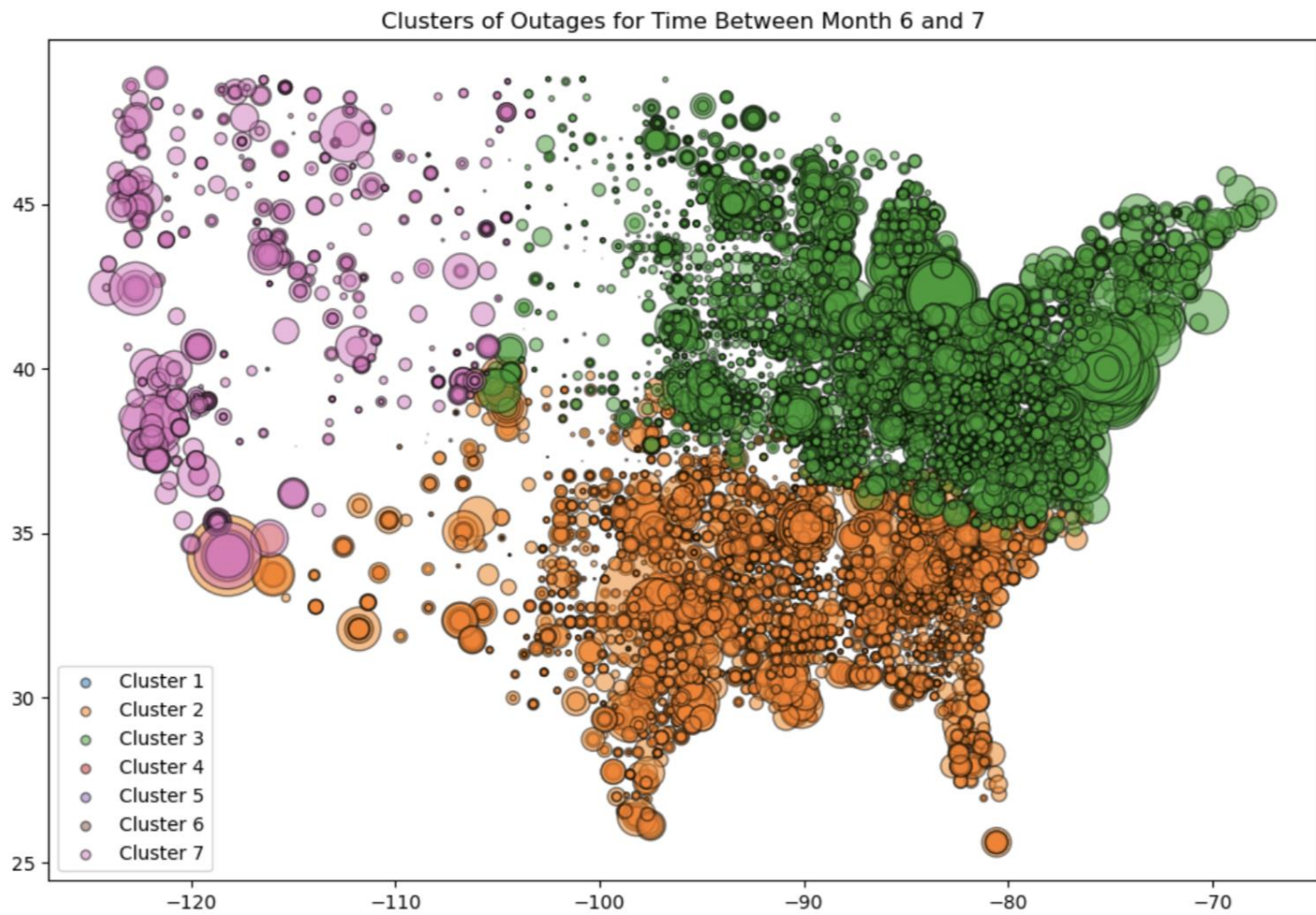


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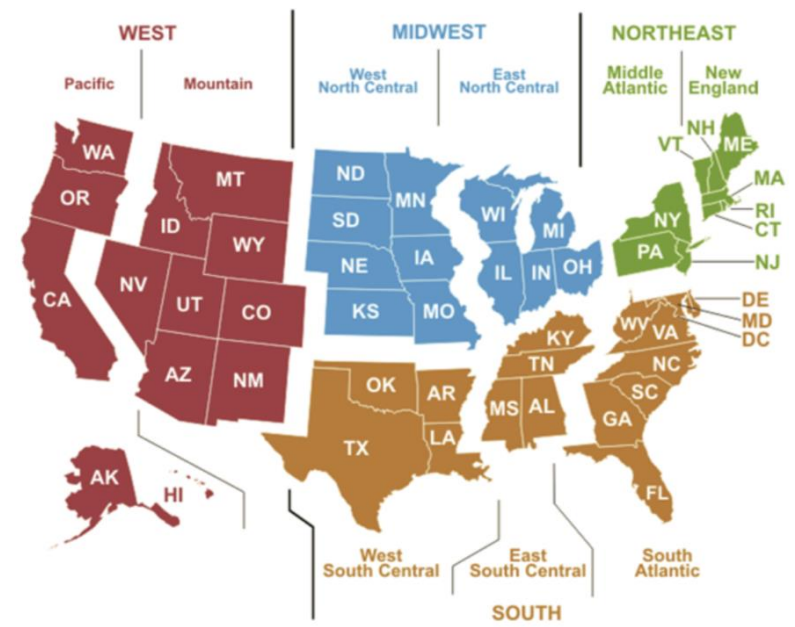
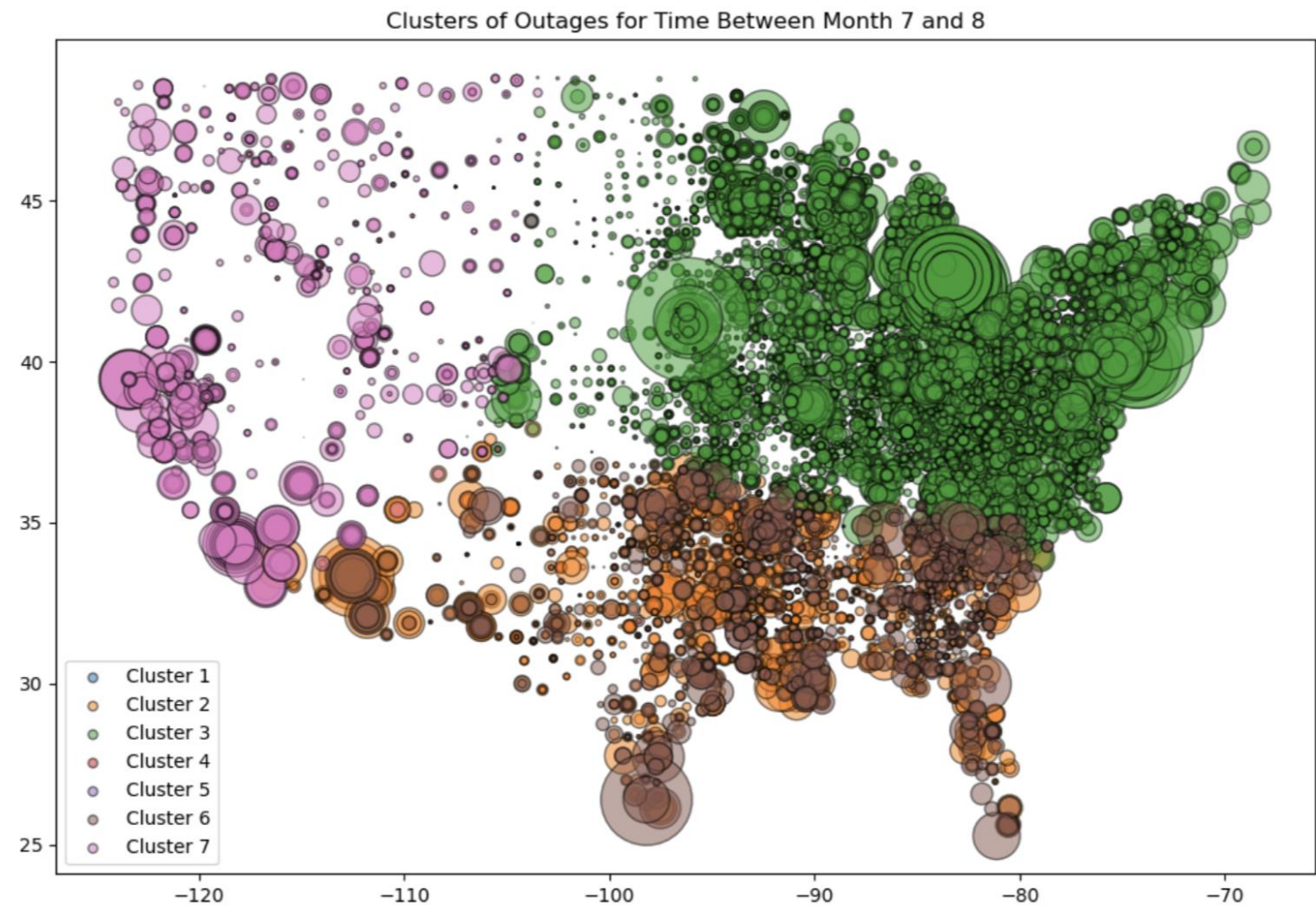


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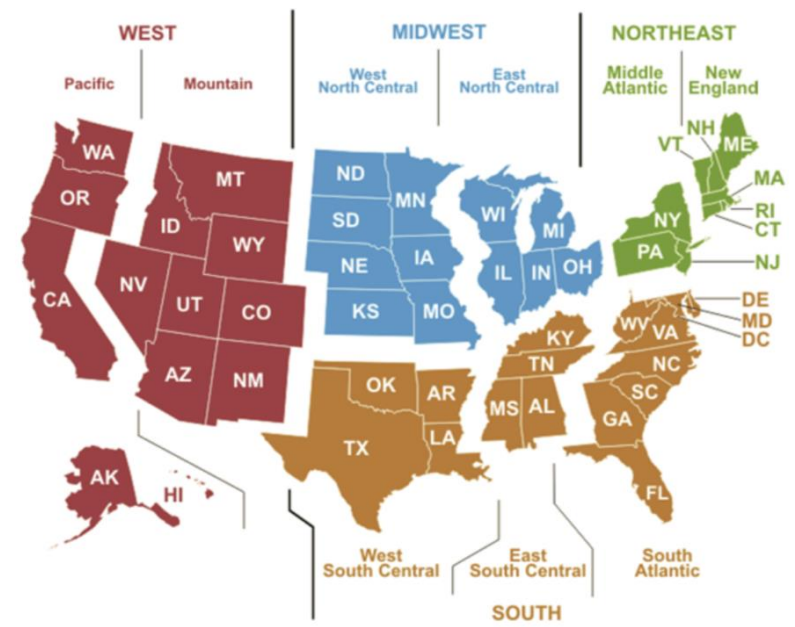
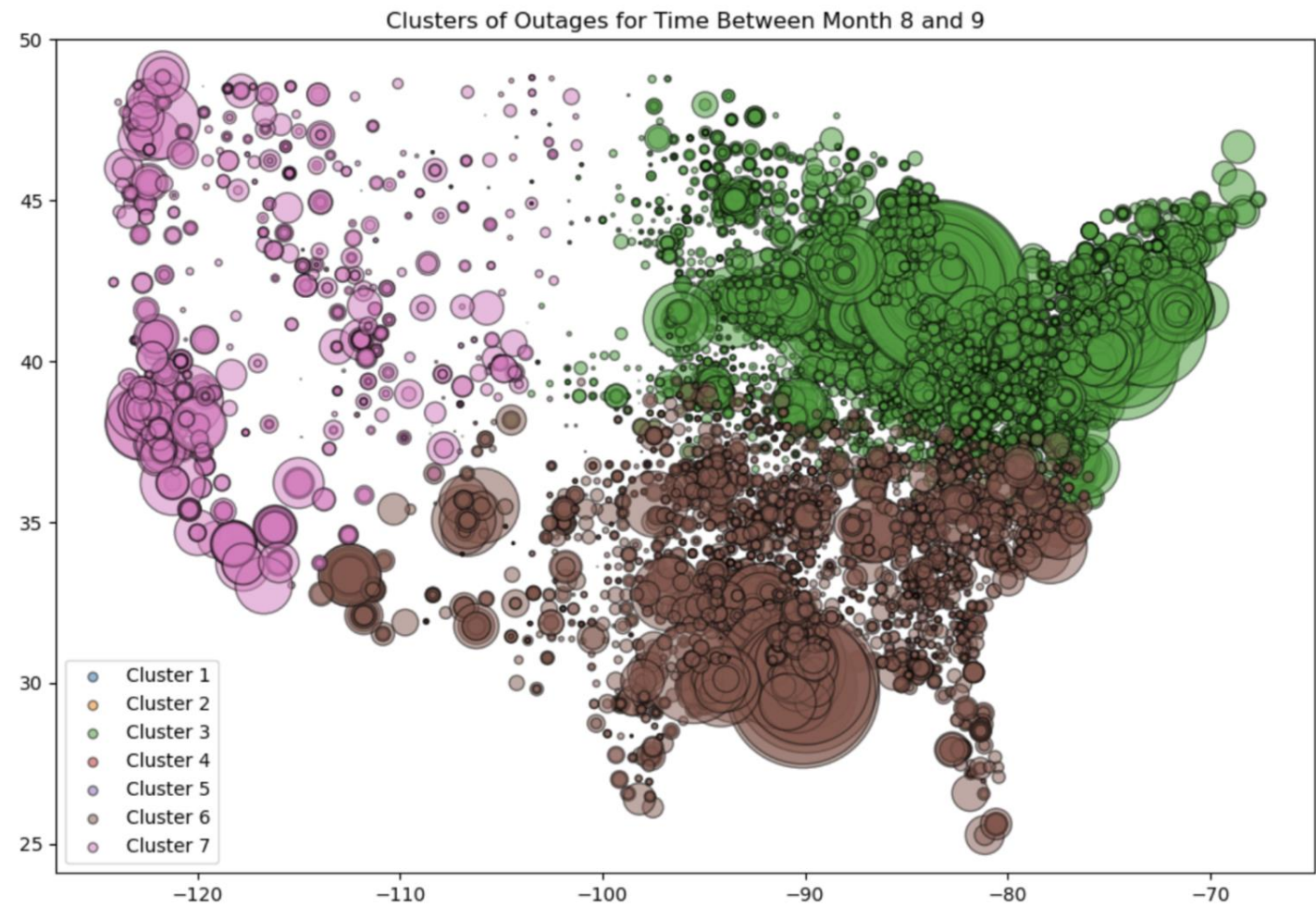
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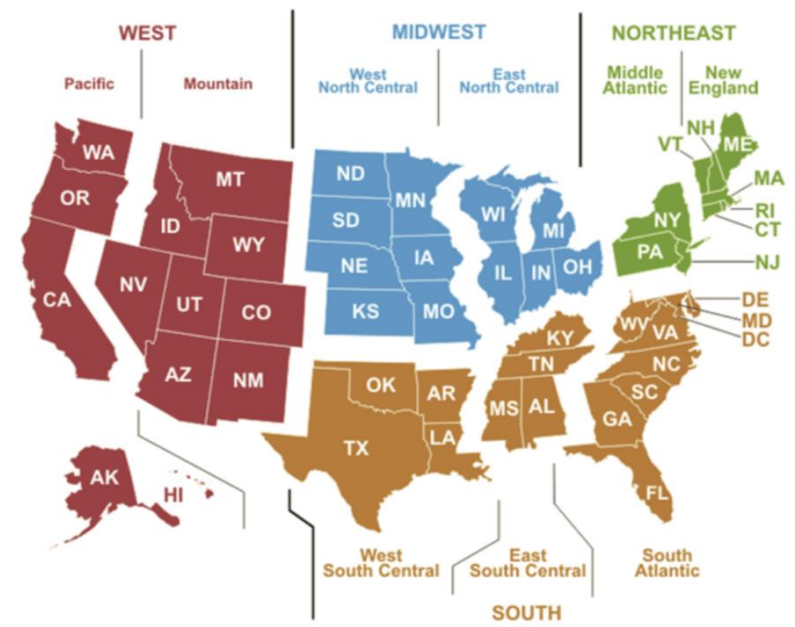
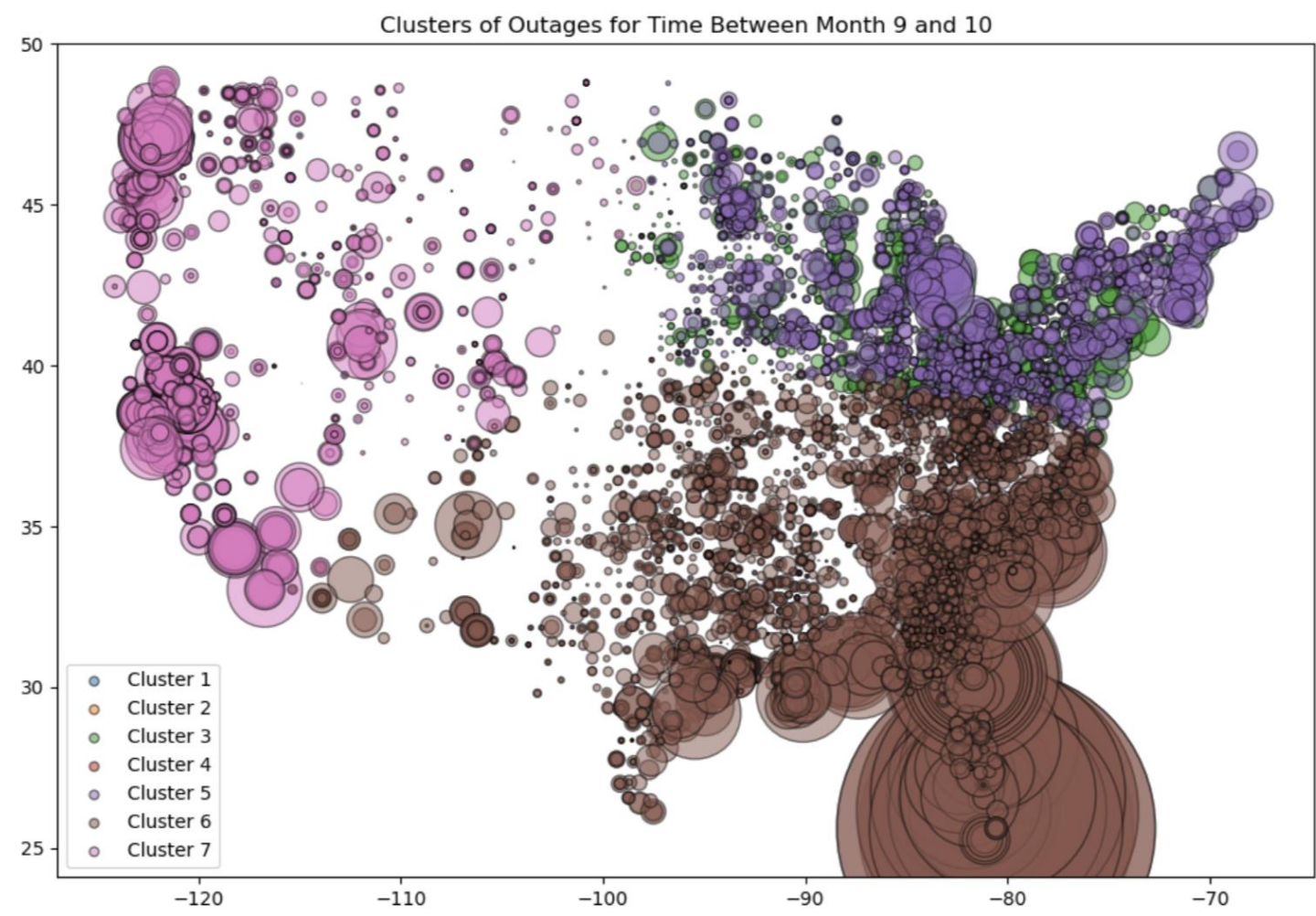


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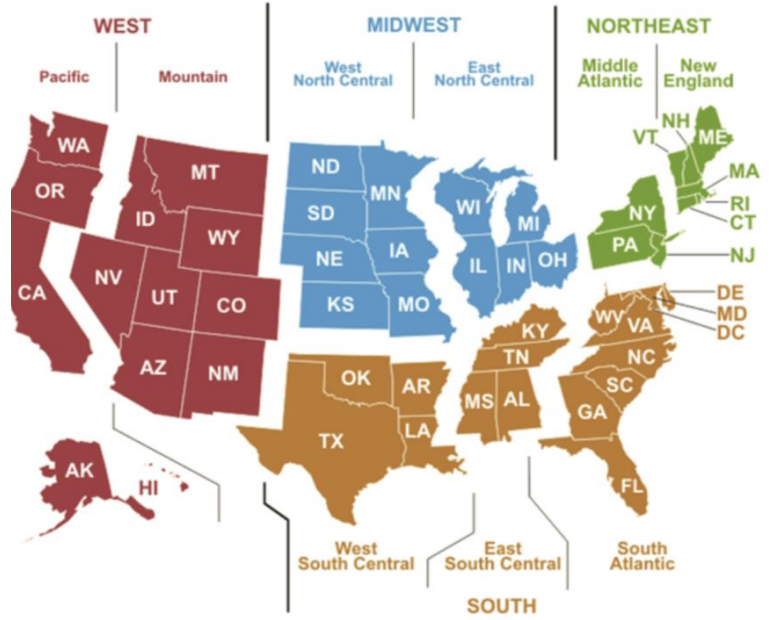
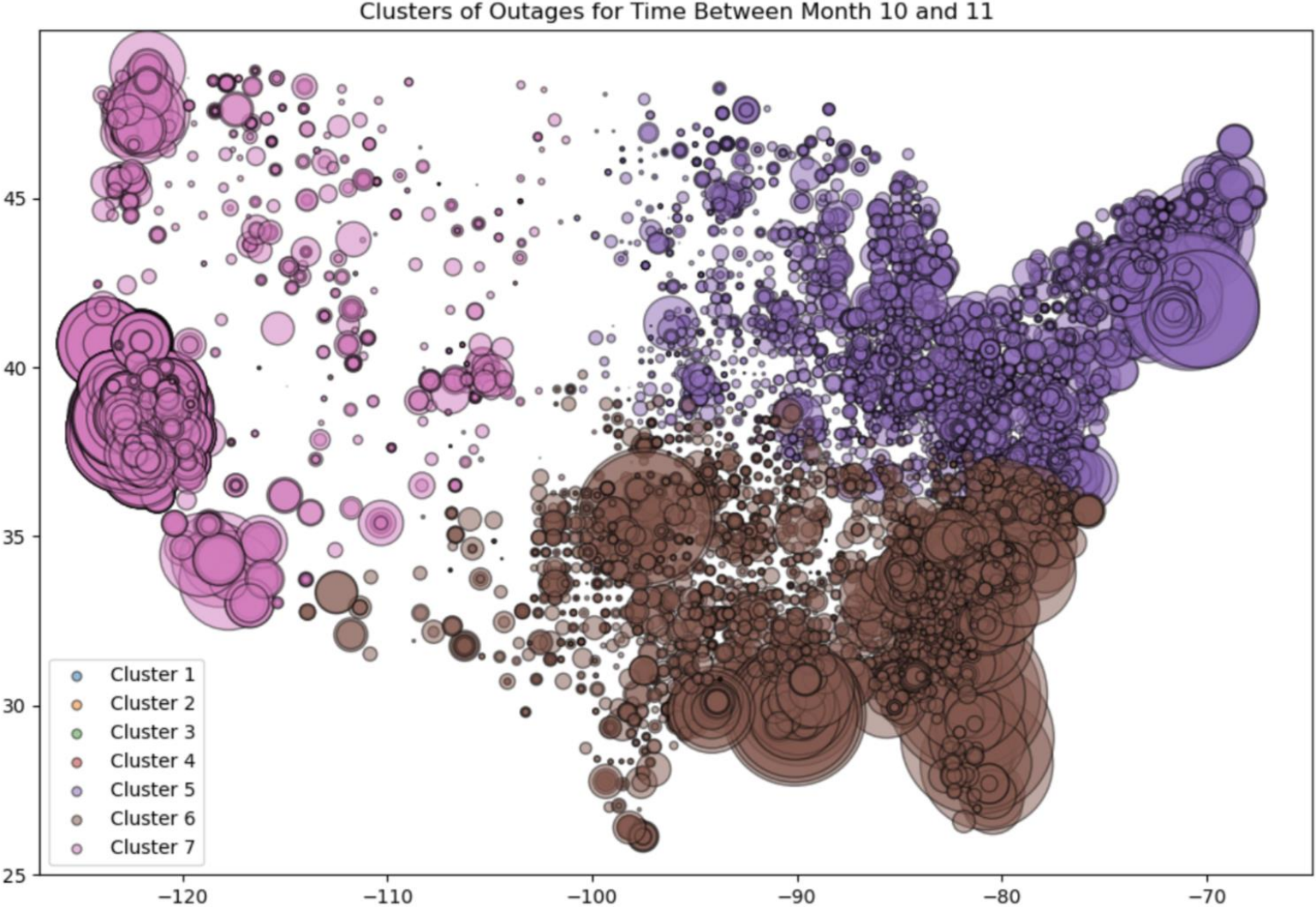


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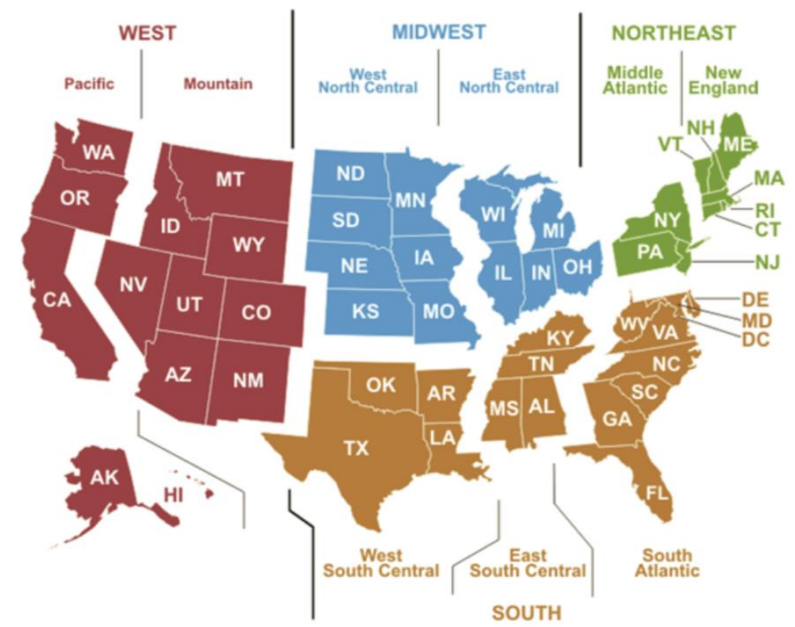
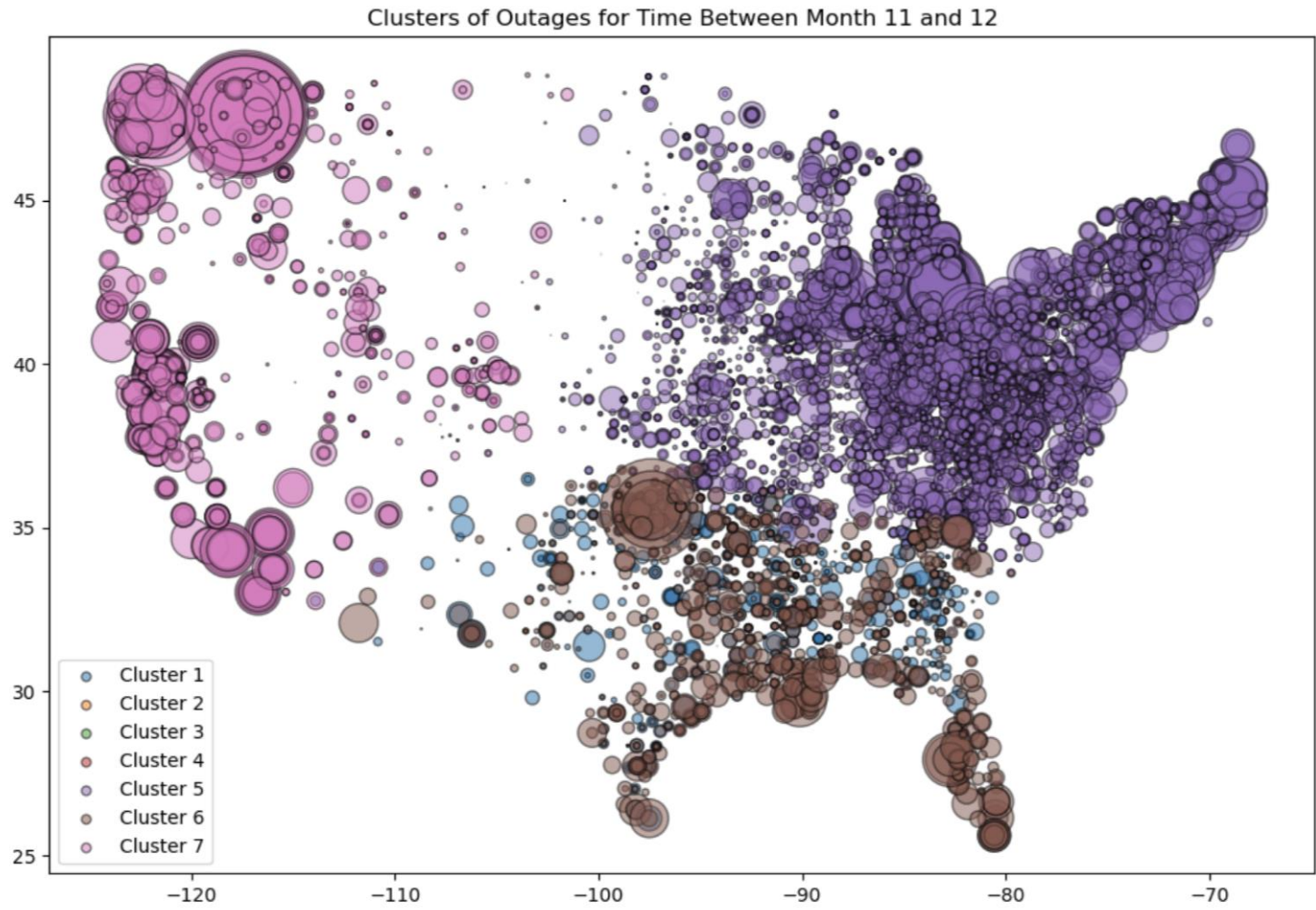
Irma	4	September 10	2017	115	Cudjoe Key Marco Island
Michael	5	October 10	2018	140	West of Mexico Beach
Ian	4	September 28	2022	130	Cayo Costa Punta Gorda

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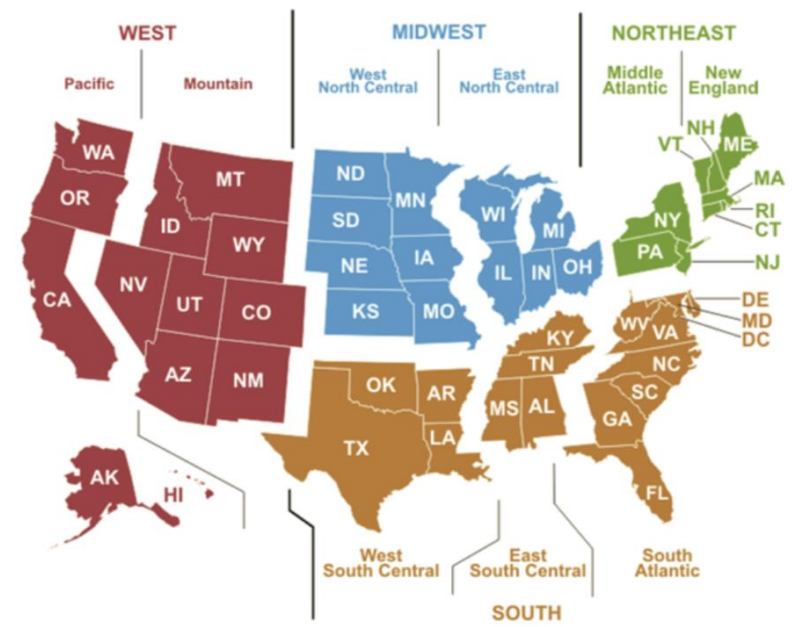
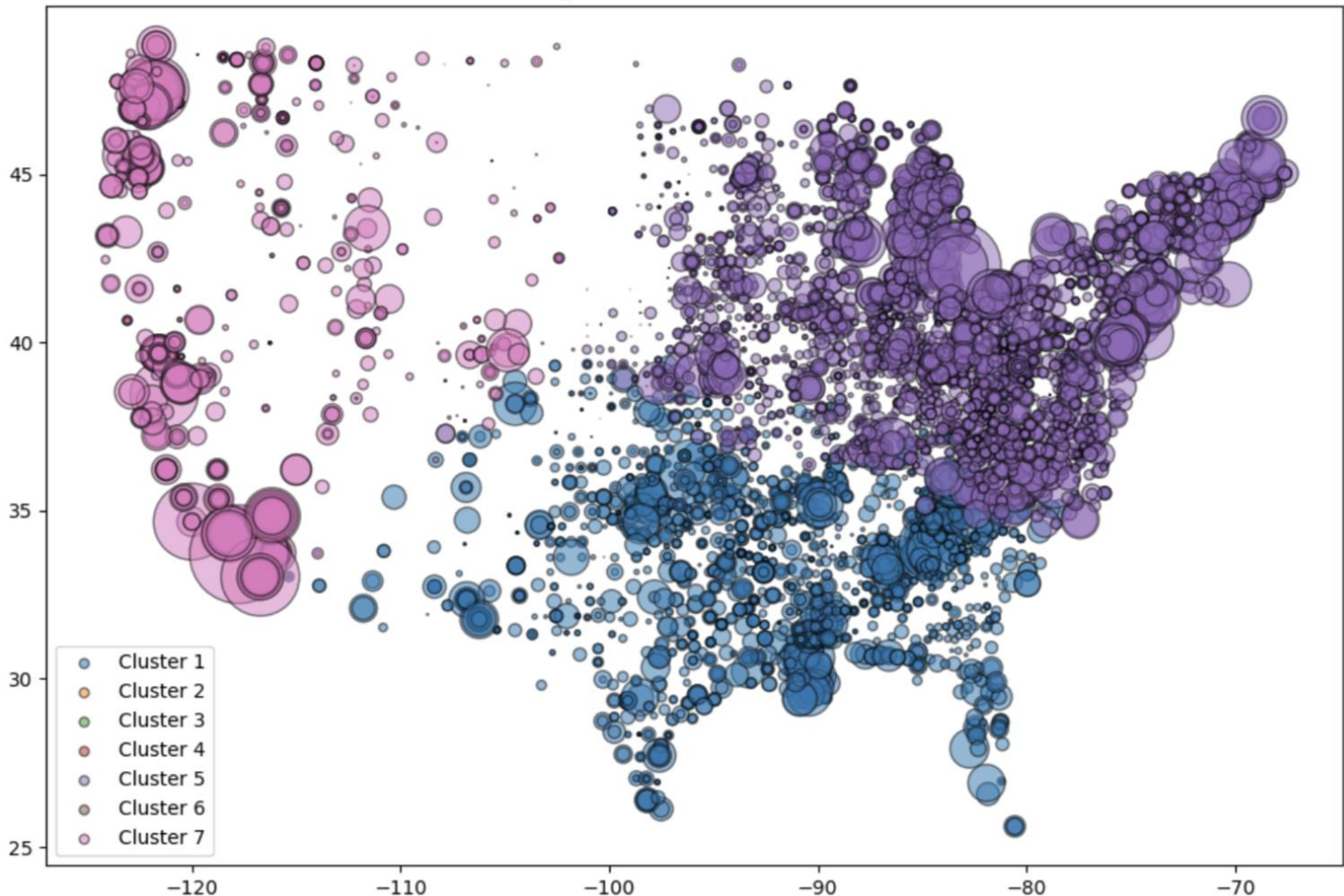
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Clusters of Outages for Time Between Month 12 and 13



Regions 1: Pacific 2: Mountain 3: West North Central 4: West South Central 5: East North Central
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Analysis

Analyze and characterize the results

This is the process of determining what story the data and results tell you. Pick a region of Questions to consider:

When are the centers of those clusters for each region?

- Use your plot slices and the results that show what time of year each cluster is centered in, to answer
- What Sevier weather happens in those the regions during the month where the cluster is centered?

What are the general characteristics of the power outages in those clusters?

- Are they generally long or short outages (large or small circles)?

Are there regions or states that mostly share clusters or that mostly do not share clusters?

- Why might that be in each case?
- Form some hypotheses about the possible relationships to:
 - Regional climate and weather
 - The time of year
 - The severity of the power outages

How well do you trust the cluster results?

K-means reference

<https://medium.com/@luo9137/k-means-clustering-clearly-explained-44746ccc3621>

<https://www.youtube.com/watch?v=4b5d3muPQmA>